

Analiza glazbenih preferencija i njihovog utjecaja na mentalno zdravlje

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Učitavanje podataka

Učitavanje podataka u data frame

```
music <- read_csv("Datasets/Music and mental health dataset.csv")
```

```
## Rows: 736 Columns: 33
## -- Column specification -----
## Delimiter: ","
## chr (26): Timestamp, Primary streaming service, While working, Instrumentali...
## dbl (7): Age, Hours per day, BPM, Anxiety, Depression, Insomnia, OCD
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
head(music, 10)
#glimpse(music)
cat("-----\n")
sum(is.na(music$BPM))
```

```
## # A tibble: 10 x 33
##   Timestamp      Age Primary streaming ser~1 'Hours per day' 'While working'
##   <chr>          <dbl> <chr>                                <dbl> <chr>
## 1 8/27/2022 19:29    18 Spotify                                3    Yes
## 2 8/27/2022 19:57    63 Pandora                                1.5  Yes
## 3 8/27/2022 21:28    18 Spotify                                4    No
## 4 8/27/2022 21:40    61 YouTube Music                        2.5  Yes
## 5 8/27/2022 21:54    18 Spotify                                4    Yes
## 6 8/27/2022 21:56    18 Spotify                                5    Yes
## 7 8/27/2022 22:00    18 YouTube Music                        3    Yes
## 8 8/27/2022 22:18    21 Spotify                                1    Yes
## 9 8/27/2022 22:33    19 Spotify                                6    Yes
## 10 8/27/2022 22:44    18 I do not use a streami~            1    Yes
## # i abbreviated name: 1: 'Primary streaming service'
## # i 28 more variables: Instrumentalist <chr>, Composer <chr>,
## #   'Fav genre' <chr>, Exploratory <chr>, 'Foreign languages' <chr>, BPM <dbl>,
## #   'Frequency [Classical]' <chr>, 'Frequency [Country]' <chr>,
## #   'Frequency [EDM]' <chr>, 'Frequency [Folk]' <chr>,
## #   'Frequency [Gospel]' <chr>, 'Frequency [Hip hop]' <chr>,
```

```
## # 'Frequency [Jazz]' <chr>, 'Frequency [K pop]' <chr>, ...
## -----
## [1] 107
```

Budući da stupac BPM ima čak 107 NA vrijednosti, umjesto da maknemo retke gdje je BPM NA odlučili smo ukloniti taj stupac jer bi 107 redaka bilo previše za izgubiti u skupu podataka s 736 vrijednosti.

Budući da ostali stupci nemaju toliko NA vrijednosti, odlučili smo izbaciti retke gdje se pojavljuje neka NA vrijednost.

Uklanjanje NA vrijednosti

```
ver1 <- music %>% select(-BPM, -Permissions) %>% na.omit()
```

Imamo puno stupaca s kategorijskim vrijednostima ili samo sa vrijednostima Da/Ne koje trebamo faktorizirati.

Faktoriziranje kategorijskih varijabli

```
ver1$`While working` <- factor(ver1$`While working`, levels = c("No", "Yes")) %>% as.numeric() - 1
ver1$Instrumentalist <- factor(ver1$Instrumentalist, levels = c("No", "Yes")) %>% as.numeric() - 1
ver1$Composer <- factor(ver1$Composer, levels = c("No", "Yes")) %>% as.numeric() - 1
ver1$Exploratory <- factor(ver1$Exploratory, levels = c("No", "Yes")) %>% as.numeric() - 1
ver1$`Foreign languages` <- factor(ver1$`Foreign languages`, levels = c("No", "Yes")) %>% as.numeric() - 1
ver1$`Music effects` <- factor(ver1$`Music effects`, levels = c("Worsen", "No effect", "Improve")) %>% as.numeric() - 1

ver2 <- ver1

ver1 <- ver1 %>%
  mutate(across(starts_with("Frequency"), ~ as.numeric(factor(., levels = c("Never", "Rarely", "Sometimes")) - 1)))
```

Sad kada su svi podaci numerizirani, možemo vidjeti neke osnovne statistike.

Sažetak statistika (summary)

```
summary(ver1)
```

```
##   Timestamp      Age   Primary streaming service Hours per day
## Length:718      Min.   :10.00 Length:718           Min.   : 0.000
## Class :character 1st Qu.:18.00 Class :character     1st Qu.: 2.000
## Mode  :character Median :21.00 Mode  :character     Median : 3.000
##                Mean  :25.18                Mean  : 3.578
##                3rd Qu.:28.00                3rd Qu.: 5.000
##                Max.   :89.00                Max.   :24.000
## While working   Instrumentalist   Composer   Fav genre
## Min.   :0.0000   Min.   :0.0000   Min.   :0.0000   Length:718
## 1st Qu.:1.0000   1st Qu.:0.0000   1st Qu.:0.0000   Class :character
```

```

## Median :1.0000 Median :0.0000 Median :0.0000 Mode :character
## Mean :0.7897 Mean :0.3189 Mean :0.1727
## 3rd Qu.:1.0000 3rd Qu.:1.0000 3rd Qu.:0.0000
## Max. :1.0000 Max. :1.0000 Max. :1.0000
## Exploratory Foreign languages Frequency [Classical] Frequency [Country]
## Min. :0.0000 Min. :0.0000 Min. :1.000 Min. :1.000
## 1st Qu.:0.0000 1st Qu.:0.0000 1st Qu.:2.000 1st Qu.:1.000
## Median :1.0000 Median :1.0000 Median :2.000 Median :2.000
## Mean :0.7159 Mean :0.5501 Mean :2.333 Mean :1.822
## 3rd Qu.:1.0000 3rd Qu.:1.0000 3rd Qu.:3.000 3rd Qu.:2.000
## Max. :1.0000 Max. :1.0000 Max. :4.000 Max. :4.000
## Frequency [EDM] Frequency [Folk] Frequency [Gospel] Frequency [Hip hop]
## Min. :1.000 Min. :1.000 Min. :1.000 Min. :1.000
## 1st Qu.:1.000 1st Qu.:1.000 1st Qu.:1.000 1st Qu.:2.000
## Median :2.000 Median :2.000 Median :1.000 Median :2.000
## Mean :2.033 Mean :2.015 Mean :1.382 Mean :2.393
## 3rd Qu.:3.000 3rd Qu.:3.000 3rd Qu.:2.000 3rd Qu.:3.000
## Max. :4.000 Max. :4.000 Max. :4.000 Max. :4.000
## Frequency [Jazz] Frequency [K pop] Frequency [Latin] Frequency [Lofi]
## Min. :1.000 Min. :1.000 Min. :1.000 Min. :1.000
## 1st Qu.:1.000 1st Qu.:1.000 1st Qu.:1.000 1st Qu.:1.000
## Median :2.000 Median :1.000 Median :1.000 Median :2.000
## Mean :2.033 Mean :1.723 Mean :1.604 Mean :2.071
## 3rd Qu.:3.000 3rd Qu.:2.000 3rd Qu.:2.000 3rd Qu.:3.000
## Max. :4.000 Max. :4.000 Max. :4.000 Max. :4.000
## Frequency [Metal] Frequency [Pop] Frequency [R&B] Frequency [Rap]
## Min. :1.000 Min. :1.000 Min. :1.000 Min. :1.000
## 1st Qu.:1.000 1st Qu.:2.000 1st Qu.:1.000 1st Qu.:1.000
## Median :2.000 Median :3.000 Median :2.000 Median :2.000
## Mean :2.227 Mean :3.038 Mean :2.258 Mean :2.344
## 3rd Qu.:3.000 3rd Qu.:4.000 3rd Qu.:3.000 3rd Qu.:3.000
## Max. :4.000 Max. :4.000 Max. :4.000 Max. :4.000
## Frequency [Rock] Frequency [Video game music] Anxiety
## Min. :1.000 Min. :1.000 Min. : 0.000
## 1st Qu.:2.250 1st Qu.:1.000 1st Qu.: 4.000
## Median :3.000 Median :2.000 Median : 6.000
## Mean :3.079 Mean :2.252 Mean : 5.838
## 3rd Qu.:4.000 3rd Qu.:3.000 3rd Qu.: 8.000
## Max. :4.000 Max. :4.000 Max. :10.000
## Depression Insomnia OCD Music effects
## Min. : 0.000 Min. : 0.000 Min. : 0.000 Min. : -1.0000
## 1st Qu.: 2.000 1st Qu.: 1.000 1st Qu.: 0.000 1st Qu.: 0.0000
## Median : 5.000 Median : 3.000 Median : 2.000 Median : 1.0000
## Mean : 4.806 Mean : 3.718 Mean : 2.635 Mean : 0.7214
## 3rd Qu.: 7.000 3rd Qu.: 6.000 3rd Qu.: 4.000 3rd Qu.: 1.0000
## Max. :10.000 Max. :10.000 Max. :10.000 Max. : 1.0000

```

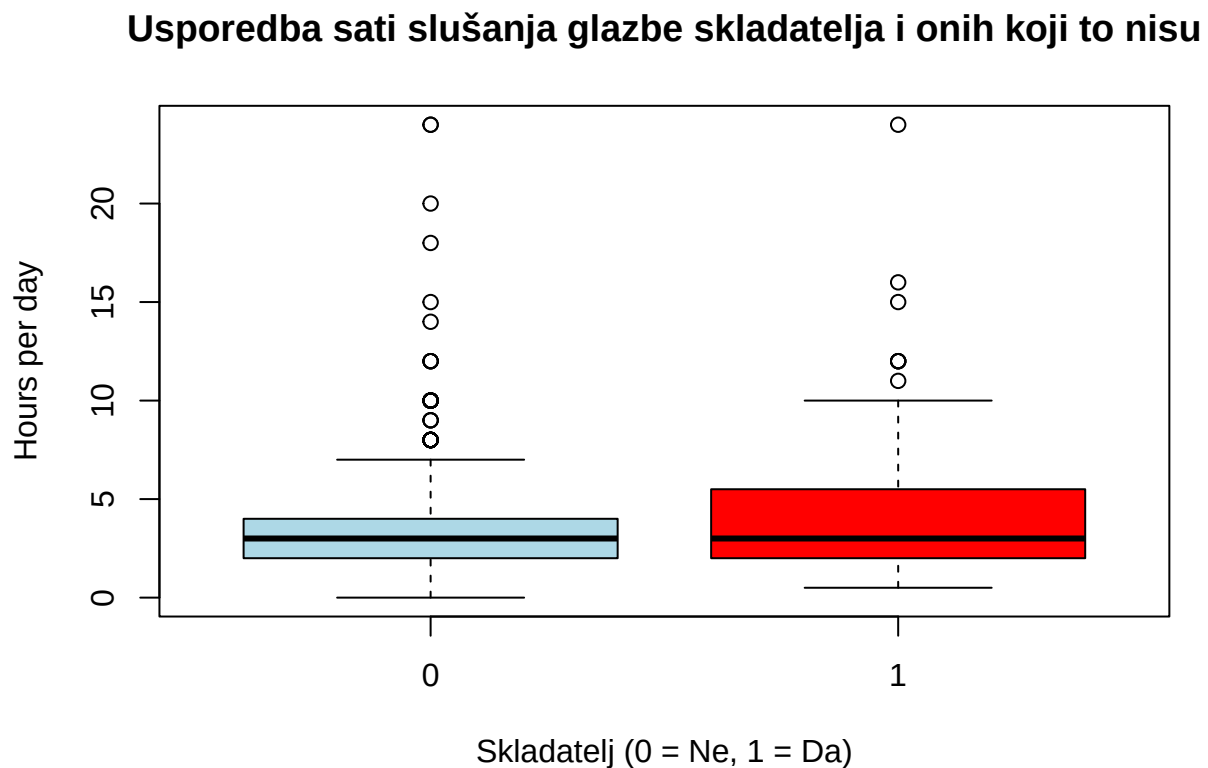
1. Slušaju li skladatelji više glazbe dnevno od drugih?

Prvo pitanje koje ćemo proučiti je slušaju li skladatelji više glazbe dnevno od onih koji nisu skladatelji. Za to trebamo podijeliti podatke na one koji imaju vrijednost 1 za stupac *Composer* i one koji to nemaju i promotriti *Hours per day*.

```
ComposerData <- ver1[ver1$Composer == 1, ]
OthersData <- ver1[ver1$Composer == 0, ]
ComposerDataHPD <- ComposerData$`Hours per day`
OthersDataHPD <- OthersData$`Hours per day`
```

Prvo ćemo promotriti kako to izgleda u pravokutnom dijagramu.

```
boxplot(`Hours per day` ~ Composer,
        data = ver1,
        main = "Usporedba sati slušanja glazbe skladatelja i onih koji to nisu",
        xlab = "Skladatelj (0 = Ne, 1 = Da)",
        ylab = "Hours per day",
        col = c("lightblue", "red"))
```

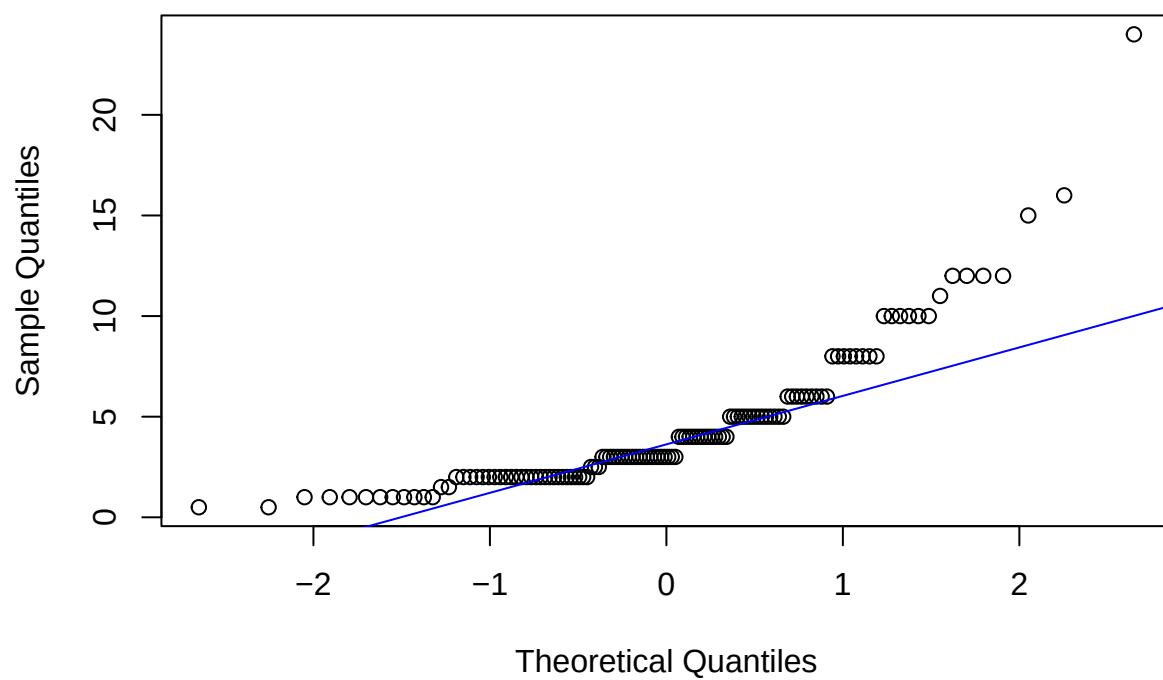


Provjera normalnosti podataka

Na prvu nam se čini da više glazbe slušaju oni koji su skladatelji, ali razlika nije velika i to moramo potvrditi. Sada trebamo promotriti ravnanju li se podaci po normalnoj razdiobi.

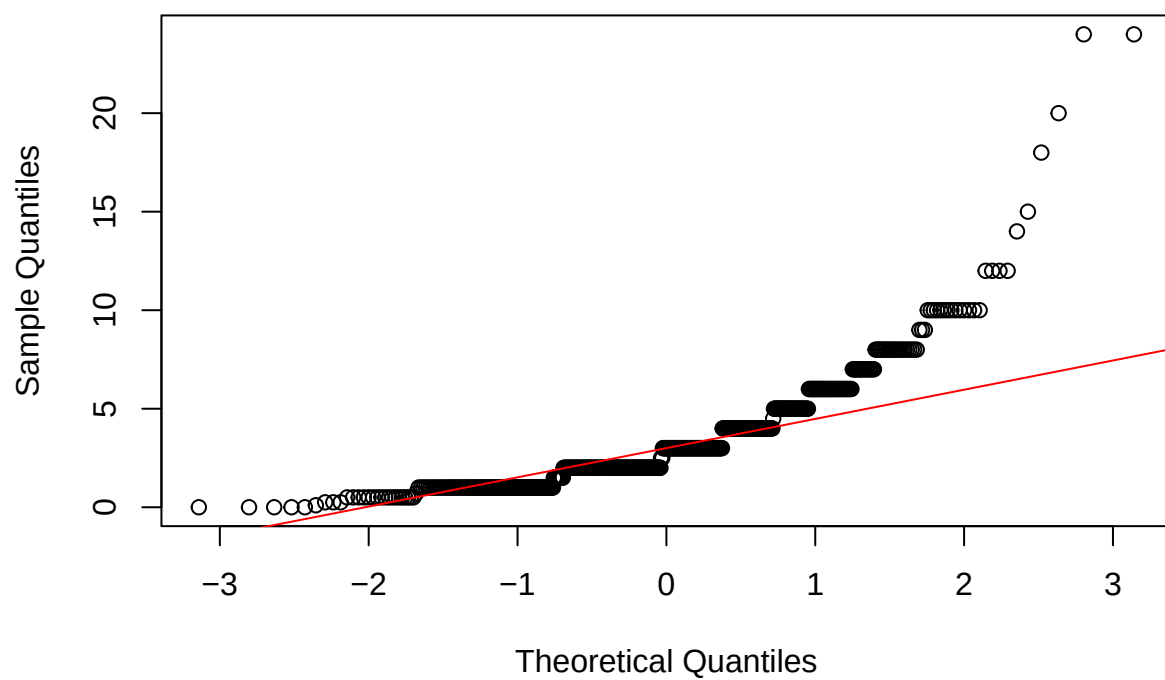
```
qqnorm(ComposerDataHPD, main = "Broj sati slušanja glazbe skladatelja")
qqline(ComposerDataHPD, col = "blue")
```

Broj sati slušanja glazbe skladatelja



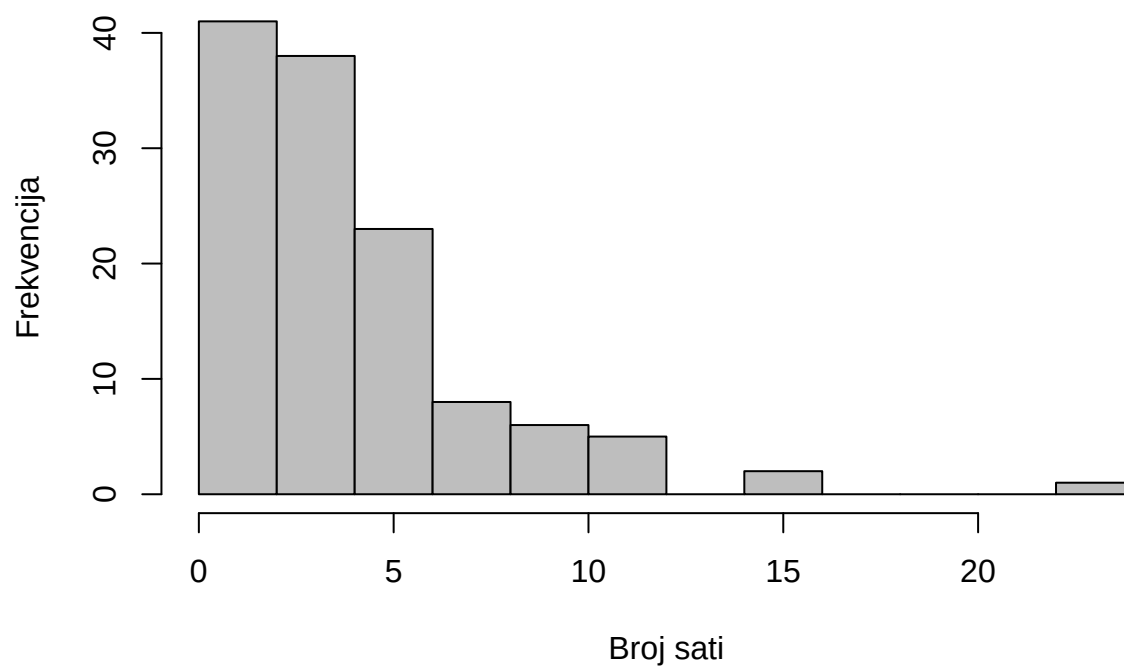
```
qqnorm(OthersDataHPD, main = "Broj sati slušanja glazbe ostalih")  
qqline(OthersDataHPD, col = "red")
```

Broj sati slušanja glazbe ostalih



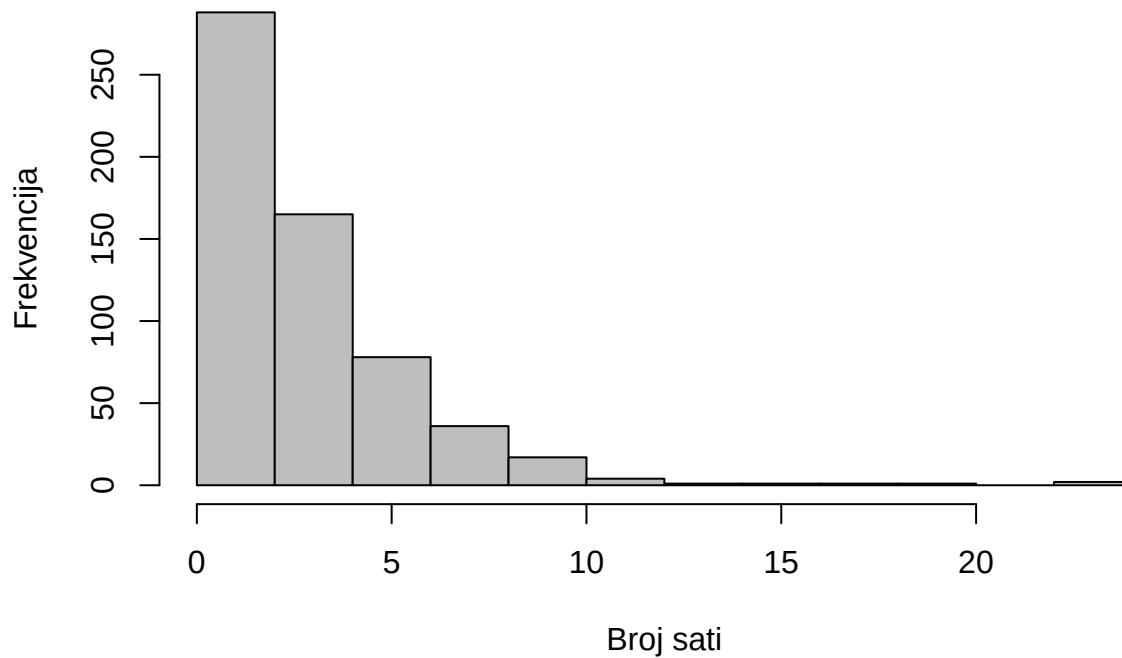
```
hist(ComposerDataHPD, main = "Slušanje glazbe skladatelja", xlab = "Broj sati", ylab = "Frekvencija", c
```

Slušanje glazbe skladatelja



```
hist(OthersDataHPD, main = "Slušanje glazbe ostalih", xlab = "Broj sati", ylab = "Frekvencija", col = "gray")
```

Slušanje glazbe ostalih



Iz qq-dijagrama i histograma možemo pretpostaviti da se podaci ne ravnaaju prema normalnoj razdiobi, ali trebamo potvrditi pomoću Kolmogorov-Smirnov test.

```
lillie.test(ComposerDataHPD)
```

```
##
##  Lilliefors (Kolmogorov-Smirnov) normality test
##
## data:  ComposerDataHPD
## D = 0.1935, p-value = 2.304e-12
```

```
lillie.test(OthersDataHPD)
```

```
##
##  Lilliefors (Kolmogorov-Smirnov) normality test
##
## data:  OthersDataHPD
## D = 0.20067, p-value < 2.2e-16
```

Iz gore navedenih testova zaključujemo da se podaci ne ravnaaju prema normalnoj razdiobi, ali zbog robustnosti t-testa mi ćemo nastaviti pod pretpostavkom da su normalni.

F test varijanci jednakosti varijanci

Sljedeće što moramo je testirati imaju li uzorci jednake varijance.


```
var.test(ComposerDataHPD, OthersDataHPD)
```

```
##
## F test to compare two variances
##
## data: ComposerDataHPD and OthersDataHPD
## F = 1.6015, num df = 123, denom df = 593, p-value = 0.0003709
## alternative hypothesis: true ratio of variances is not equal to 1
## 95 percent confidence interval:
##  1.231075 2.136478
## sample estimates:
## ratio of variances
##           1.601462
```

Budući da je p vrijednost manja od $\alpha = 0.05$, možemo odbaciti hipotezu da su varijance jednake u korist alternativne hipoteze da nisu.

t-test za uzorke s nejednakim i nepoznatim varijancama

Sada provodimo t-test za uzorke s nejednakim i nepoznatim varijancama.

```
t.test(ComposerDataHPD, OthersDataHPD, alt = "greater", var.equal = FALSE)
```

```
##
## Welch Two Sample t-test
##
## data: ComposerDataHPD and OthersDataHPD
## t = 3.1965, df = 156.6, p-value = 0.0008415
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
##  0.5304846      Inf
## sample estimates:
## mean of x mean of y
##  4.487903  3.388131
```

Prema t-testu za $\alpha = 0.05$ možemo odbaciti hipotezu H_0 da su im srednje vrijednosti jednake u korist alternativne hipoteze H_1 da je srednja vrijednost broja sati u kojima skladatelji slušaju glazbu veća od srednje vrijednosti u kojima oni koji nisu skladatelji slušaju glazbu.

Mann-Whitney__Wilcoxonov test

Budući da su podaci neparametarski, još ćemo provesti i Mann-Whitney-Wilcoxonov test, alternativni test t-testu, ali za neparametarske podatke.

```
wilcox.test(ComposerDataHPD, OthersDataHPD, alternative = "greater")
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: ComposerDataHPD and OthersDataHPD
## W = 44631, p-value = 8.53e-05
## alternative hypothesis: true location shift is greater than 0
```

Još jednom potvrđujemo da skladatelji slušaju više glazbe od ostalih, odnosno prema Mann-Whitney-Wilcoxonov testu odbacujemo hipotezu H_0 da su distribucije količine slušanja glazbe jednake za skladatelje i ostale u korist hipoteze H_1 da skladatelji slušaju više glazbe.

2. Korelacije između frekvencija slušanja različitih glazbenih žanrova i samoprocijenjenih razina mentalnih poremećaja

Korelacija između žanrova i mentalnih poremećaja

Idemo pokazati korelacije između slušanja nekog glazbenog žanra i samprocijenjene razine mentalnog poremećaja.

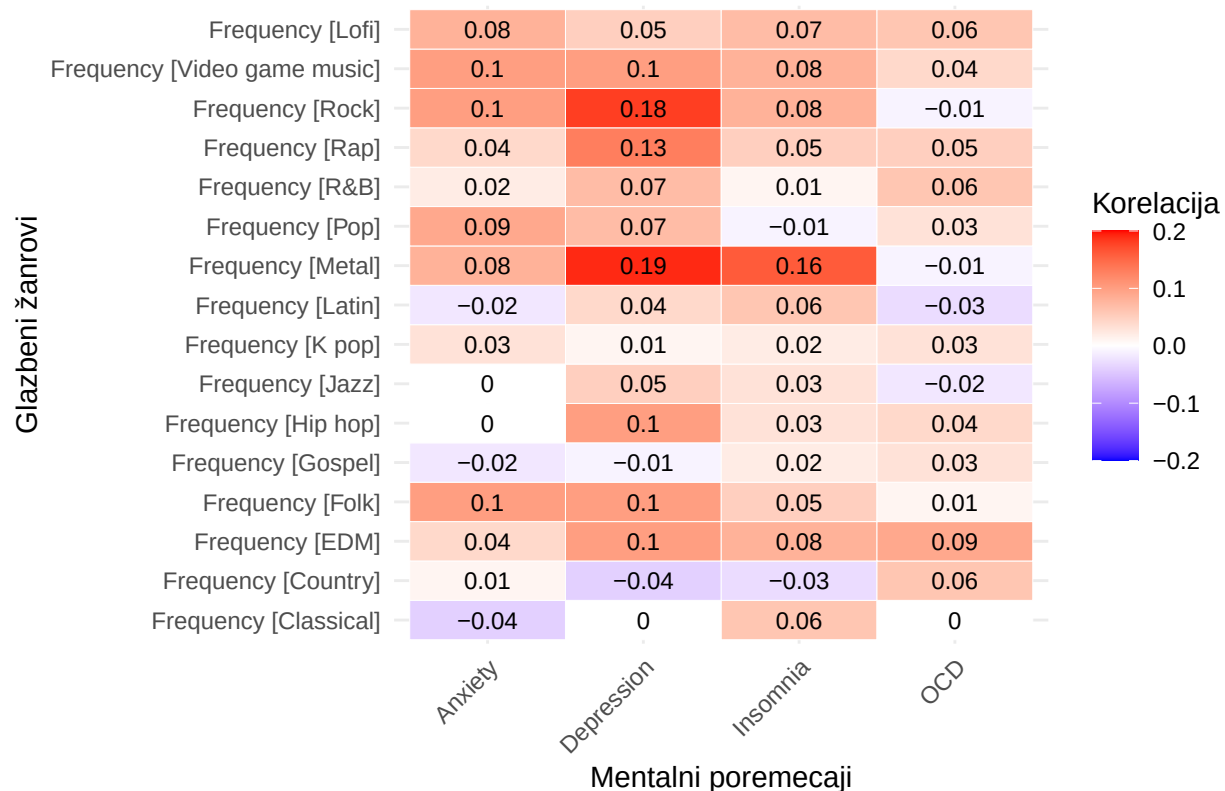
```
genres <- ver1 %>% select(`Frequency [Classical]`, `Frequency [Country]`, `Frequency [EDM]`, `Frequency [Folk]`, `Frequency [Gospel]`, `Frequency [Jazz]`, `Frequency [Latin]`, `Frequency [Pop]`, `Frequency [R&B]`, `Frequency [Rock]`, `Frequency [Soul]`, `Frequency [Symphony]`, `Frequency [World]`)
mentalHealth <- ver1 %>% select(Anxiety, Depression, Insomnia, OCD)

correlation_matrix <- cor(genres, mentalHealth, method = "spearman") %>% round(2)

heatmap_data <- melt(correlation_matrix)

ggplot(heatmap_data, aes(Var2, Var1, fill = value)) +
  geom_tile(color = "white") +
  scale_fill_gradient2(low = "blue", high = "red", mid = "white", midpoint = 0,
    limit = c(-0.2, 0.2), space = "Lab",
    name = "Korelacija") +
  geom_text(aes(label = round(value, 2)), color = "black", size = 3) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(title = "Korelacija između glazbenih žanrova i mentalnih poremećaja",
    x = "Mentalni poremećaji",
    y = "Glazbeni žanrovi")
```

Korelacija između glazbenih žanrova i mentalnih poremećaja



Na prvu djeluje da nema neke velike korelacije između određenog žanra i mentalnog poremećaja budući da je najveća korelacija 0.19.

Spearmanov test

Sad ćemo testirati je li korelacija između frekvencije slušanja metala i samoprocijenjene razine depresije statistički značajna. Koristit ćemo Spearmanov neparametarski test iz razloga što on ne zahtjeva nikakvu distribuciju, za razliku od Personovog testa koji zahtjeva normalnu distribuciju.

```
cor.test(ver1$`Frequency [Metal]`, ver1$Depression, method = "spearman")
```

```
## Warning in cor.test.default(ver1$`Frequency [Metal]`, ver1$Depression, method =
## "spearman"): Cannot compute exact p-value with ties
```

```
##
## Spearman's rank correlation rho
##
## data: ver1$`Frequency [Metal]` and ver1$Depression
## S = 50089794, p-value = 3.864e-07
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
## rho
## 0.1880524
```

Prema Spearmanov testu na značajnosti od $\alpha = 0.05$ korelacija između frekvencije slušanja metal žanra i depresije je statistički značajna. $\rho = 0.188$ označava korelaciju koja nije velika, ali ipak je statistički značajna. Porastom frekvencije slušanja metala blago će porasti samoprocijenjena razina depresije.

3. Postoji li povezanost između korištenog streaming servisa i sklonosti istraživanju?

Sljedeće istraživačko pitanje na koje tražimo odgovor je postoji li povezanost između korištenog streaming servisa i sklonosti istraživanju. Za dobiti odgovor na ovo pitanje trebat ćemo provesti χ^2 test nezavisnosti za kategorijske podatke.

Promotrimo koji se sve streaming servisi nalaze u našem skupu podataka.

```
ver1$`Primary streaming service` <- gsub("I do not use a streaming service.", "No streaming service", ver1$`Primary streaming service`)
```

```
##
##           Apple Music   No streaming service   Other streaming service
##                50                69                49
##           Pandora           Spotify           YouTube Music
##                10                450                90
```

Napravimo kontingencijsku tablicu.

```
tbl = table(ver1[ver1$`Primary streaming service` == "Apple Music" | ver1$`Primary streaming service` == "No streaming service" | ver1$`Primary streaming service` == "Other streaming service"],
            ver1[ver1$`Primary streaming service` == "Apple Music" | ver1$`Primary streaming service` == "No streaming service" | ver1$`Primary streaming service` == "Other streaming service"])
tbl
```

```
##
##                0    1
##   Apple Music    11  39
##   No streaming service  44  25
##   Other streaming service  20  29
##   Pandora         3    7
##   Spotify        101 349
##   YouTube Music   25  65
```

Podsjetimo, u stupcu *Exploratory*, 0 znači *Ne* i 1 znači *Da*. Nadogradimo tablicu da nam pišu sume vrijednosti.

```
addmargins(tbl)
added_margins_tbl <- addmargins(tbl)
```

```
##
##                0    1 Sum
##   Apple Music    11  39  50
##   No streaming service  44  25  69
##   Other streaming service  20  29  49
##   Pandora         3    7  10
##   Spotify        101 349 450
##   YouTube Music   25  65  90
##   Sum            204 514 718
```

Izračunajmo očekivane frekvencije. Pretpostavka korištenja *chisq.test()* je da su sve očekivane frekvencije veće ili jednake od 5.

```
for (col_names in colnames(added_margins_tbl)){
  for (row_names in rownames(added_margins_tbl)){
    if (!(row_names == 'Sum' | col_names == 'Sum') ){
      cat('Očekivane frekvencije za razred ', col_names, '-', row_names, ': ', (added_margins_tbl[row_names,
```

```
## Očekivane frekvencije za razred 0 - Apple Music : 14.20613
## Očekivane frekvencije za razred 0 - No streaming service : 19.60446
## Očekivane frekvencije za razred 0 - Other streaming service : 13.92201
## Očekivane frekvencije za razred 0 - Pandora : 2.841226
## Očekivane frekvencije za razred 0 - Spotify : 127.8552
## Očekivane frekvencije za razred 0 - YouTube Music : 25.57103
## Očekivane frekvencije za razred 1 - Apple Music : 35.79387
## Očekivane frekvencije za razred 1 - No streaming service : 49.39554
## Očekivane frekvencije za razred 1 - Other streaming service : 35.07799
## Očekivane frekvencije za razred 1 - Pandora : 7.158774
## Očekivane frekvencije za razred 1 - Spotify : 322.1448
## Očekivane frekvencije za razred 1 - YouTube Music : 64.42897
```

Budući da je očekivana frekvencija za razred 0 - Pandora manja od 5 ne možemo koristiti klasičan *chisq.test()*, već moramo koristiti Fisherov egzaktan test.

Fisherov egzaktan test

Početne hipoteze:

H0: ne postoji povezanost između korištenog streaming servisa i sklonosti istraživanju.

H1: postoji povezanost između korištenog streaming servisa i sklonosti istraživanju.

```
fisher.test(tbl, simulate.p.value = TRUE, B = 10000)
```

```
##
## Fisher's Exact Test for Count Data with simulated p-value (based on
## 10000 replicates)
##
## data:  tbl
## p-value = 9.999e-05
## alternative hypothesis: two.sided
```

Na značajnosti od $\alpha = 0.05$ možemo odbaciti hipotezu H0 u korist hipoteze H1 koja tvrdi da postoji povezanost između korištenog streaming servisa i sklonosti istraživanju.

4. Ovisnost prosječnog broja sati slušanja o različitim preferiranim žanrovima glazbe

Obrada podataka

```
data_prime <- (ver1 %>%
  rename(
    favorite = `Fav genre`,
    hours = `Hours per day`
  ))[c("hours", "favorite")]

data_prime$favorite = factor(data_prime$favorite,
                             levels = c("Latin", "Lofi", "Gospel", "Rock", "Video game music", "Jazz",
                                           "Other", "Other", "Other", "Rock", "VGM", "Jazz", "RnB", "Kpop"),
                             labels = c("Other", "Other", "Other", "Rock", "VGM", "Jazz", "RnB", "Kpop"))

summary(data_prime)
```

```
##      hours      favorite
##  Min.   : 0.000   Rock    :184
##  1st Qu.: 2.000   Pop     :114
##  Median : 3.000   Metal   : 87
##  Mean   : 3.578   Classical: 51
##  3rd Qu.: 5.000   VGM     : 43
##  Max.   :24.000   EDM     : 36
##                      (Other) :203
```

Provjera normalnosti i homogenosti varijanci

Prvi test kojeg želimo isprobati jest ANOVA test koji je parametarski test za testiranje hipoteze o jednakosti srednjih vrijednosti između više kategorija podataka.

No kako bi proveli ANOVA test moramo znati da podatci dolaze iz normalnih distribucija otprilike jednakih varijanaca. Zato koristimo sljedeće testove:

```
print(paste("Testing: data"))

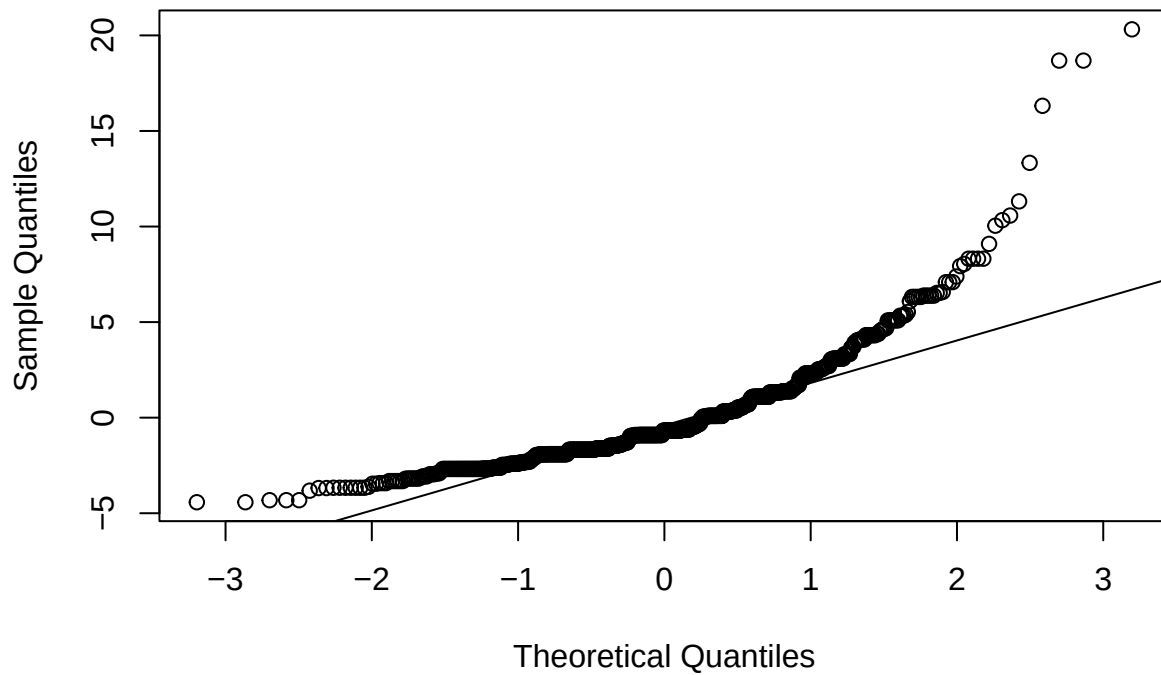
print(
  data %>%
  group_by(data$favorite) %>%
  summarise(Count = n(), Var = var(data$hours), AvgHours = mean(data$hours))
)
print("Overall test results:")

print(paste("Total mean:", mean(data$hours)))

#ANOVA pokušaj
anova_result <- aov(data$hours ~ data$favorite)
print(anova_result)

#QQ plot
qqnorm(residuals(anova_result))
qqline(residuals(anova_result))
```

Normal Q-Q Plot



```
#lillie test o normalnosti podataka
print(lillie.test(data$hours))

# Barlett i Levenin test jednakosti varijanca
print(bartlett.test(data$hours ~ data$favorite))
print(leveneTest(data$hours ~ data$favorite))
```

```
## [1] "Testing: data"
## # A tibble: 14 x 4
##   'data$favorite' Count    Var AvgHours
##   <fct>          <int> <dbl>   <dbl>
## 1 Other             18  9.11    3.58
## 2 Rock            184  9.11    3.58
## 3 VGM              43  9.11    3.58
## 4 Jazz             20  9.11    3.58
## 5 RnB              35  9.11    3.58
## 6 Kpop             21  9.11    3.58
## 7 Country          24  9.11    3.58
## 8 EDM              36  9.11    3.58
## 9 HipHop           34  9.11    3.58
## 10 Pop            114  9.11    3.58
## 11 Rap             22  9.11    3.58
## 12 Classical       51  9.11    3.58
## 13 Metal           87  9.11    3.58
## 14 Folk            29  9.11    3.58
## [1] "Overall test results:"
```

```
## [1] "Total mean: 3.57806406685237"
## Call:
## aov(formula = data$hours ~ data$favorite)
##
## Terms:
##              data$favorite Residuals
## Sum of Squares      286.177  6243.535
## Deg. of Freedom       13      704
##
## Residual standard error: 2.978029
## Estimated effects may be unbalanced
##
## Lilliefors (Kolmogorov-Smirnov) normality test
##
## data: data$hours
## D = 0.2013, p-value < 2.2e-16
##
##
## Bartlett test of homogeneity of variances
##
## data: data$hours by data$favorite
## Bartlett's K-squared = 116.28, df = 13, p-value < 2.2e-16
##
## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value Pr(>F)
## group 13  2.1155 0.01172 *
##      704
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
print(paste("Testing: trimmed_data"))

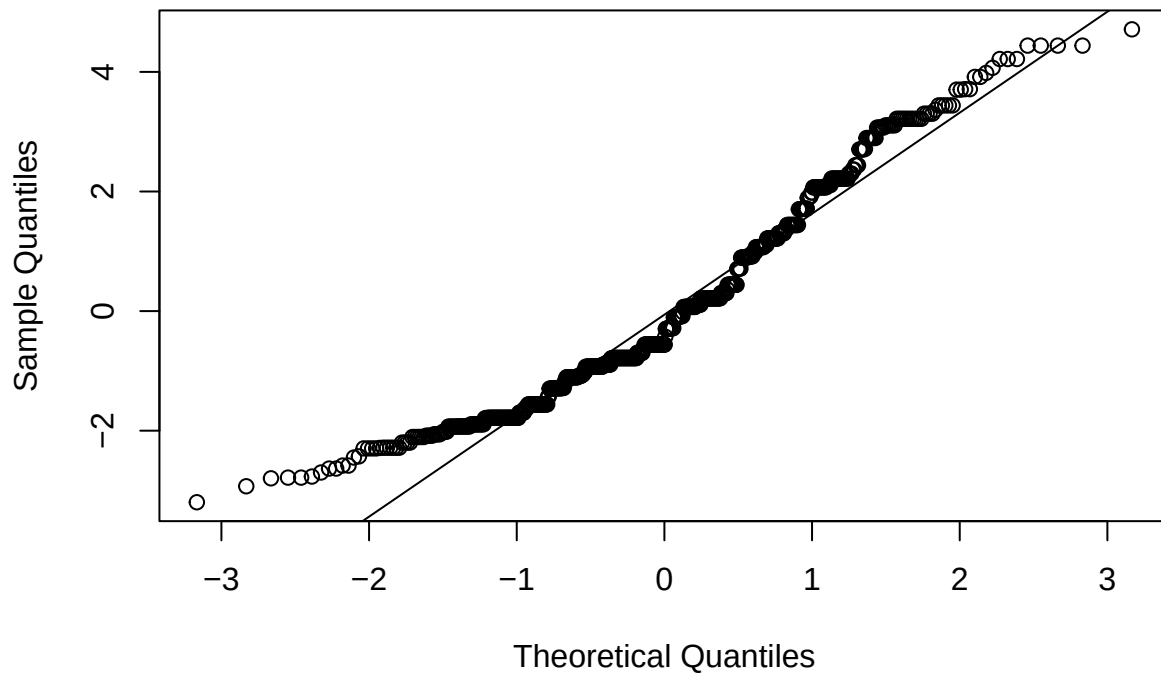
print(
  trimmed_data %>%
  group_by(trimmed_data$favorite) %>%
  summarise(Count = n(), Var = var(trimmed_data$hours), AvgHours = mean(trimmed_data$hours))
)
print("Overall test results:")

print(paste("Total mean:", mean(trimmed_data$hours)))

#ANOVA pokusaj
anova_result <- aov(trimmed_data$hours ~ trimmed_data$favorite)
print(anova_result)

#QQ plot
qqnorm(residuals(anova_result))
qqline(residuals(anova_result))
```


Normal Q-Q Plot



```
#lillie test o normalnosti podataka
print(lillie.test(trimmed_data$hours))

# Barlett i Levenin test jednakosti varijanca
print(bartlett.test(trimmed_data$hours ~ trimmed_data$favorite))
print(leveneTest(trimmed_data$hours ~ trimmed_data$favorite))
```

```
## [1] "Testing: trimmed_data"
## # A tibble: 14 x 4
##   'trimmed_data$favorite' Count   Var AvgHours
##   <fct>                 <int> <dbl>   <dbl>
## 1 Other                   15  2.80    2.83
## 2 Rock                   164  2.80    2.83
## 3 VGM                     39  2.80    2.83
## 4 Jazz                    15  2.80    2.83
## 5 RnB                     33  2.80    2.83
## 6 Kpop                    19  2.80    2.83
## 7 Country                 22  2.80    2.83
## 8 EDM                     28  2.80    2.83
## 9 HipHop                  30  2.80    2.83
## 10 Pop                   108  2.80    2.83
## 11 Rap                    19  2.80    2.83
## 12 Classical              49  2.80    2.83
## 13 Metal                  78  2.80    2.83
## 14 Folk                   29  2.80    2.83
## [1] "Overall test results:"
```

```
## [1] "Total mean: 2.82878086419753"
## Call:
## aov(formula = trimmed_data$hours ~ trimmed_data$favorite)
##
## Terms:
##             trimmed_data$favorite Residuals
## Sum of Squares             44.0611 1766.5796
## Deg. of Freedom              13      634
##
## Residual standard error: 1.669252
## Estimated effects may be unbalanced
##
## Lilliefors (Kolmogorov-Smirnov) normality test
##
## data:  trimmed_data$hours
## D = 0.19757, p-value < 2.2e-16
##
##
## Bartlett test of homogeneity of variances
##
## data:  trimmed_data$hours by trimmed_data$favorite
## Bartlett's K-squared = 9.0388, df = 13, p-value = 0.77
##
## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value Pr(>F)
## group  13   0.881 0.5732
##      634
```

```
datasets <- list(data = data, trimmed_data = trimmed_data)
h1 <- ggplot(datasets$data[datasets$data$favorite == "Rock", ], aes(x = hours)) +
  geom_histogram(binwidth = 1, fill = "lightblue", color = "black") +
  labs(title = "Data - Rock Frequency", x = "Hours", y = "Count") +
  theme_classic()

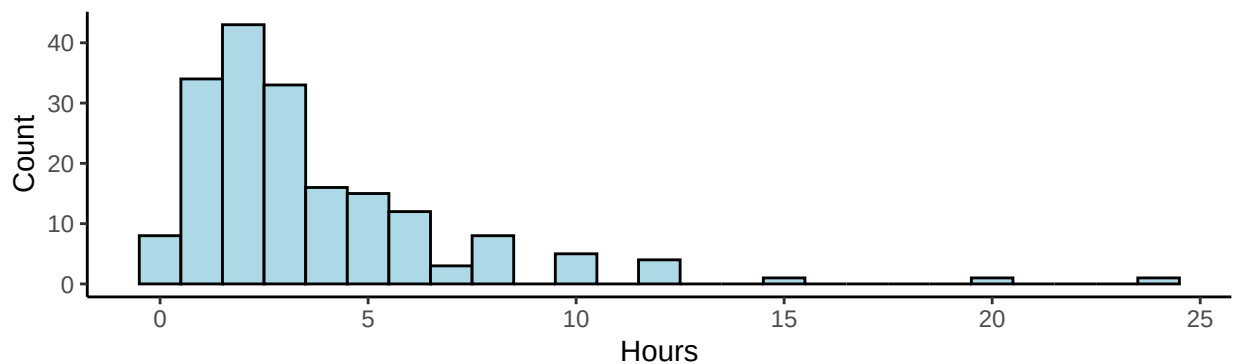
print(paste("Test results for Rock in Data"))
print(lillie.test(datasets$data$hours[datasets$data$favorite == "Rock"]))

h2 <- ggplot(datasets$trimmed_data[datasets$trimmed_data$favorite == "Rock", ], aes(x = hours)) +
  geom_histogram(binwidth = 1, fill = "lightblue", color = "black") +
  labs(title = "Trimmed Data - Rock Frequency", x = "Hours", y = "Count") +
  theme_classic()

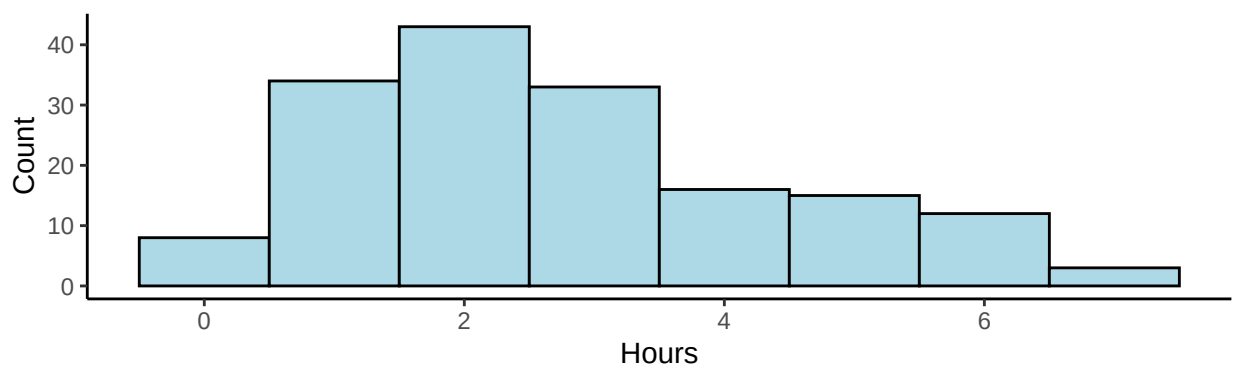
print(paste("Test results for Rock in Trimmed Data"))
print(lillie.test(datasets$trimmed_data$hours[datasets$trimmed_data$favorite == "Rock"]))

grid.arrange(h1, h2)
```

Data – Rock Frequency



Trimmed Data – Rock Frequency



```
## [1] "Test results for Rock in Data"
##
##  Lilliefors (Kolmogorov-Smirnov) normality test
##
## data:  datasets$data$hours[datasets$data$favorite == "Rock"]
## D = 0.22277, p-value < 2.2e-16
##
## [1] "Test results for Rock in Trimmed Data"
##
##  Lilliefors (Kolmogorov-Smirnov) normality test
##
## data:  datasets$trimmed_data$hours[datasets$trimmed_data$favorite == "Rock"]
## D = 0.18687, p-value = 2.441e-15
```

```
h1 <- ggplot(datasets$data[datasets$data$favorite == "Rap", ], aes(x = hours)) +
  geom_histogram(binwidth = 1, fill = "lightblue", color = "black") +
  labs(title = "Data - Rap Frequency", x = "Hours", y = "Count") +
  theme_classic()

print(paste("Test results for Rap in Data"))
print(lillie.test(datasets$data$hours[datasets$data$favorite == "Rap"]))

h2 <- ggplot(datasets$trimmed_data[datasets$trimmed_data$favorite == "Rap", ], aes(x = hours)) +
  geom_histogram(binwidth = 1, fill = "lightblue", color = "black") +
  labs(title = "Trimmed Data - Rap Frequency", x = "Hours", y = "Count") +
```

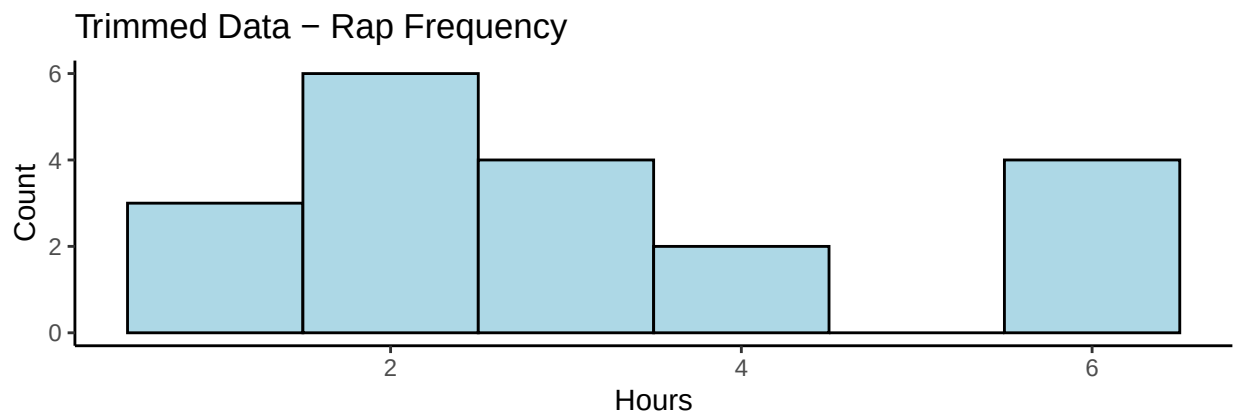
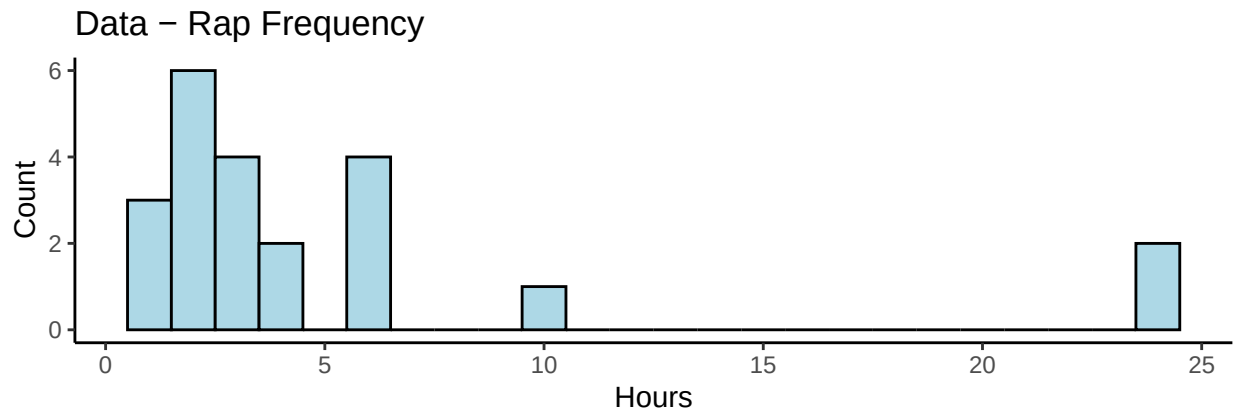
```

theme_classic()

print(paste("Test results for Rap in Trimmed Data"))
print(lillie.test(datasets$trimmed_data$hours[datasets$trimmed_data$favorite == "Rap"]))

grid.arrange(h1, h2)

```



```

## [1] "Test results for Rap in Data"
##
##  Lilliefors (Kolmogorov-Smirnov) normality test
##
## data:  datasets$data$hours[datasets$data$favorite == "Rap"]
## D = 0.32143, p-value = 2.266e-06
##
## [1] "Test results for Rap in Trimmed Data"
##
##  Lilliefors (Kolmogorov-Smirnov) normality test
##
## data:  datasets$trimmed_data$hours[datasets$trimmed_data$favorite == "Rap"]
## D = 0.20862, p-value = 0.029

```

```

h1 <- ggplot(datasets$data[datasets$data$favorite == "HipHop", ], aes(x = hours)) +
  geom_histogram(binwidth = 1, fill = "lightblue", color = "black") +
  labs(title = "Data - HipHop Frequency", x = "Hours", y = "Count") +
  theme_classic()

```

```

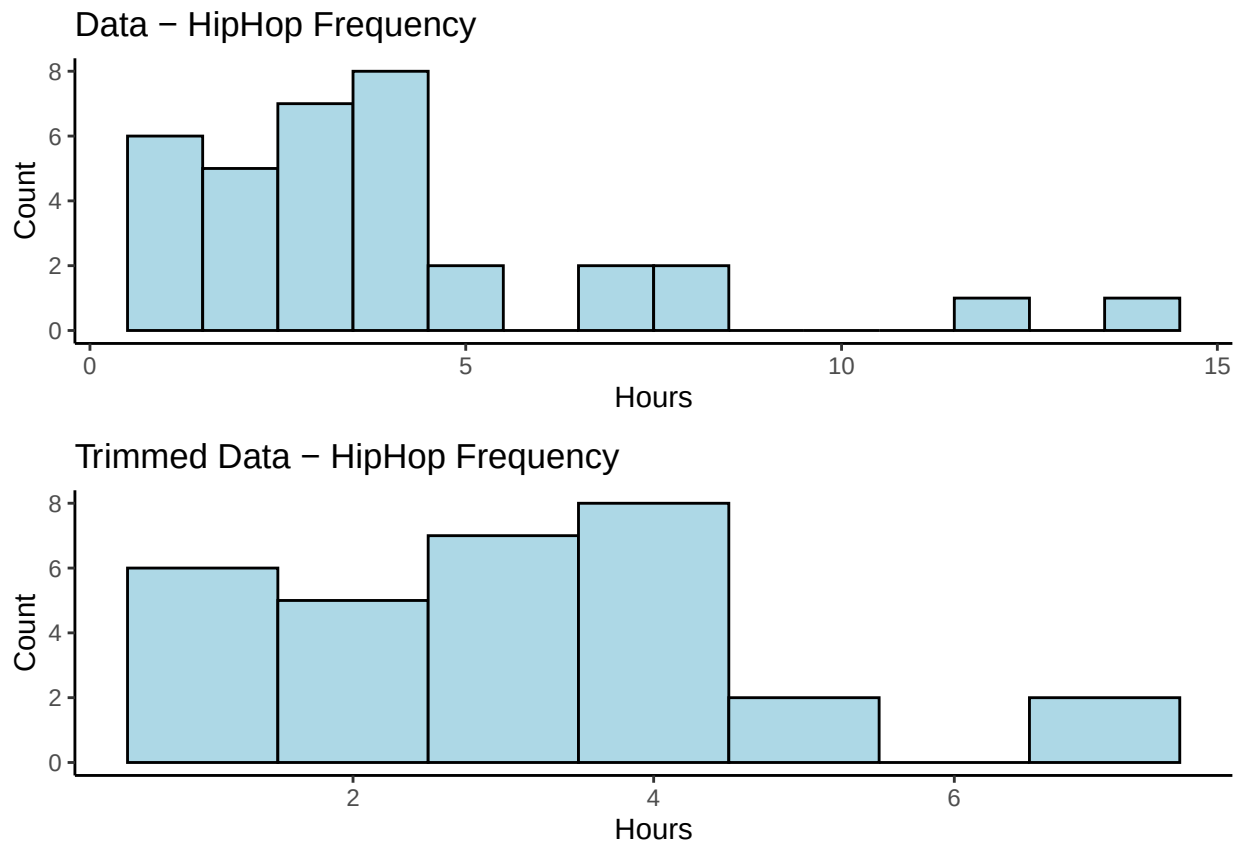
print(paste("Test results for HipHop in Data"))
print(lillie.test(datasets$data$hours[datasets$data$favorite == "HipHop"]))

h2 <- ggplot(datasets$trimmed_data[datasets$trimmed_data$favorite == "HipHop", ], aes(x = hours)) +
  geom_histogram(binwidth = 1, fill = "lightblue", color = "black") +
  labs(title = "Trimmed Data - HipHop Frequency", x = "Hours", y = "Count") +
  theme_classic()

print(paste("Test results for HipHop in Trimmed Data"))
print(lillie.test(datasets$trimmed_data$hours[datasets$trimmed_data$favorite == "HipHop"]))

grid.arrange(h1, h2)

```



```

## [1] "Test results for HipHop in Data"
##
## Lilliefors (Kolmogorov-Smirnov) normality test
##
## data: datasets$data$hours[datasets$data$favorite == "HipHop"]
## D = 0.25888, p-value = 3.775e-06
##
## [1] "Test results for HipHop in Trimmed Data"
##
## Lilliefors (Kolmogorov-Smirnov) normality test
##
## data: datasets$trimmed_data$hours[datasets$trimmed_data$favorite == "HipHop"]

```

```
## D = 0.15663, p-value = 0.05837
```

Ovi rezultati na razini značajnosti od 1% upućuju na to da se ne radi o normalno distribuiranim podacima, te vidimo da na individualnoj razini većina grupa doista značajno odstupa od normalne distribucije.

Također vidimo da kod nepodrezenih podataka homogenost varijance također ne prolazi testove.

No pomnijom analizom vidimo da je uzrok najvjerojatnije Rap kategorija koja jedina značajno odstupa od ostalih.

Relativno robustne metode vjerojatno će pretrpjeti ovakvo odstupanje, no normalnost je ovdje ipak značajno prekršena čak i kod podrezenih podataka, te ANOVA test nije primjenjiv.

Kruskal-Wallis Test

Umjesto njega upotrebljavamo neparametarski Kruskal-Wallis test koji ne zahtjeva pretpostavku normalnosti i testira nul hipotezu o jednakosti medijana među kategorijama podataka.

```
kruskal.test(AvgTime ~ FavGenre)

kruskal.test(AvgTime_t ~ FavGenre_t)

##
##  Kruskal-Wallis rank sum test
##
## data:  AvgTime by FavGenre
## Kruskal-Wallis chi-squared = 24.237, df = 13, p-value = 0.02902
##
##
##  Kruskal-Wallis rank sum test
##
## data:  AvgTime_t by FavGenre_t
## Kruskal-Wallis chi-squared = 17.642, df = 13, p-value = 0.1716
```

Rezultat nam ukazuje na to da se barem jedna grupa razlikuje od drugih po medijanu u nepodrezanom datasetu na razini značajnosti od 5%, no da bismo bili sigurni da ovo nije greška, moramo detaljnije analizirati podatke.

Dunn Test

Kako bi provjerili o kojim se skupinama radi, provodimo Dunn test. On nam za svaki par žanrova govori koliko se značajno razlikuju po srednjoj vrijednosti.

```
dunnTest(AvgTime ~ FavGenre)

## Dunn (1964) Kruskal-Wallis multiple comparison

##  p-values adjusted with the Holm method.

##           Comparison      Z      P.unadj      P.adj
## 1  Classical - Country -0.619034188 0.535893837 1.00000000
## 2    Classical - EDM -1.803716483 0.071275764 1.00000000
## 3    Country - EDM -0.908472452 0.363628662 1.00000000
```

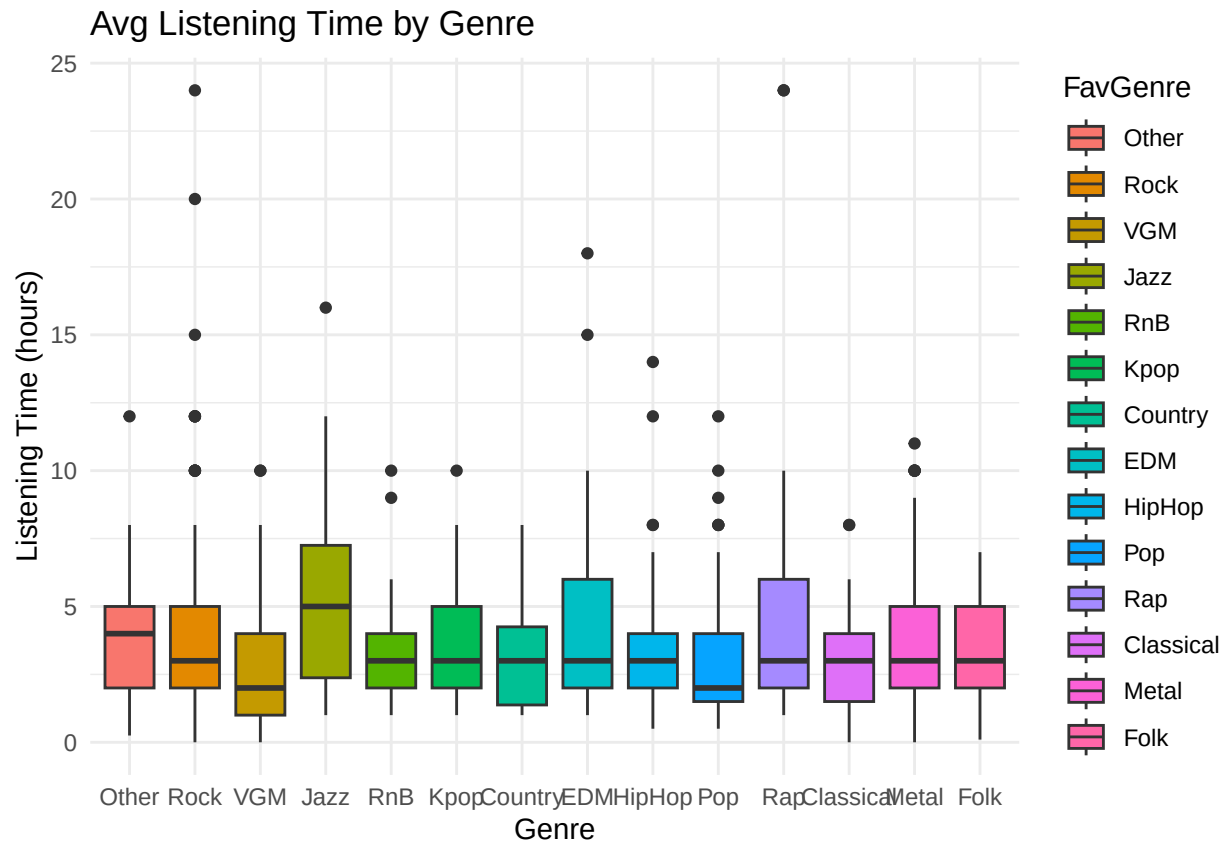
## 4	Classical - Folk	-0.704846484	0.480905777	1.00000000
## 5	Country - Folk	-0.038757031	0.969084103	1.00000000
## 6	EDM - Folk	0.916591356	0.359356816	1.00000000
## 7	Classical - HipHop	-1.546849697	0.121899493	1.00000000
## 8	Country - HipHop	-0.709829939	0.477809601	1.00000000
## 9	EDM - HipHop	0.209744603	0.833867012	1.00000000
## 10	Folk - HipHop	-0.706360820	0.479963780	1.00000000
## 11	Classical - Jazz	-2.735684354	0.006225071	0.55403128
## 12	Country - Jazz	-1.877792689	0.060409531	1.00000000
## 13	EDM - Jazz	-1.180144077	0.237942917	1.00000000
## 14	Folk - Jazz	-1.919206186	0.054958245	1.00000000
## 15	HipHop - Jazz	-1.345934111	0.178323767	1.00000000
## 16	Classical - Kpop	-0.769835738	0.441397337	1.00000000
## 17	Country - Kpop	-0.155184912	0.876675543	1.00000000
## 18	EDM - Kpop	0.703000069	0.482055707	1.00000000
## 19	Folk - Kpop	-0.124506241	0.900914459	1.00000000
## 20	HipHop - Kpop	0.514779611	0.606707034	1.00000000
## 21	Jazz - Kpop	1.671229106	0.094676433	1.00000000
## 22	Classical - Metal	-1.496251743	0.134588065	1.00000000
## 23	Country - Metal	-0.479870044	0.631319803	1.00000000
## 24	EDM - Metal	0.649747330	0.515855446	1.00000000
## 25	Folk - Metal	-0.466121269	0.641128700	1.00000000
## 26	HipHop - Metal	0.388635853	0.697545541	1.00000000
## 27	Jazz - Metal	1.846468918	0.064824148	1.00000000
## 28	Kpop - Metal	-0.264348307	0.791511540	1.00000000
## 29	Classical - Other	-1.258122037	0.208347625	1.00000000
## 30	Country - Other	-0.614782869	0.538698124	1.00000000
## 31	EDM - Other	0.165276698	0.868726199	1.00000000
## 32	Folk - Other	-0.603195500	0.546378641	1.00000000
## 33	HipHop - Other	-0.008397342	0.993299969	0.99329997
## 34	Jazz - Other	1.159882064	0.246096826	1.00000000
## 35	Kpop - Other	-0.452422916	0.650964341	1.00000000
## 36	Metal - Other	-0.313008197	0.754274434	1.00000000
## 37	Classical - Pop	0.427976378	0.668668319	1.00000000
## 38	Country - Pop	1.003324257	0.315704436	1.00000000
## 39	EDM - Pop	2.430881987	0.015062120	1.00000000
## 40	Folk - Pop	1.134867528	0.256430827	1.00000000
## 41	HipHop - Pop	2.121610504	0.033870460	1.00000000
## 42	Jazz - Pop	3.274613169	0.001058067	0.09628414
## 43	Kpop - Pop	1.144164715	0.252555333	1.00000000
## 44	Metal - Pop	2.360041321	0.018272900	1.00000000
## 45	Other - Pop	1.644231548	0.100128383	1.00000000
## 46	Classical - Rap	-1.441922378	0.149324272	1.00000000
## 47	Country - Rap	-0.726927446	0.467270403	1.00000000
## 48	EDM - Rap	0.091796216	0.926859951	1.00000000
## 49	Folk - Rap	-0.721061401	0.470871738	1.00000000
## 50	HipHop - Rap	-0.092529384	0.926277446	1.00000000
## 51	Jazz - Rap	1.145685797	0.251925183	1.00000000
## 52	Kpop - Rap	-0.551304366	0.581425046	1.00000000
## 53	Metal - Rap	-0.435469195	0.663221894	1.00000000
## 54	Other - Rap	-0.071958355	0.942635050	1.00000000
## 55	Pop - Rap	-1.889045515	0.058885730	1.00000000
## 56	Classical - RnB	-1.307506618	0.191040709	1.00000000
## 57	Country - RnB	-0.504711411	0.613761545	1.00000000

## 58	EDM - RnB	0.445035690	0.656293962	1.000000000
## 59	Folk - RnB	-0.490095635	0.624066227	1.000000000
## 60	HipHop - RnB	0.230416041	0.817768498	1.000000000
## 61	Jazz - RnB	1.551051513	0.120889340	1.000000000
## 62	Kpop - RnB	-0.316602644	0.751545129	1.000000000
## 63	Metal - RnB	-0.115500353	0.908048541	1.000000000
## 64	Other - RnB	0.199730873	0.841691067	1.000000000
## 65	Pop - RnB	-1.858233162	0.063135911	1.000000000
## 66	Rap - RnB	0.296978677	0.766482789	1.000000000
## 67	Classical - Rock	-1.010037973	0.312477098	1.000000000
## 68	Country - Rock	-0.030426026	0.975727289	1.000000000
## 69	EDM - Rock	1.277415420	0.201455624	1.000000000
## 70	Folk - Rock	0.020479891	0.983660553	1.000000000
## 71	HipHop - Rock	0.978405284	0.327873916	1.000000000
## 72	Jazz - Rock	2.386648461	0.017002741	1.000000000
## 73	Kpop - Rock	0.172649335	0.862927071	1.000000000
## 74	Metal - Rock	0.799610670	0.423936404	1.000000000
## 75	Other - Rock	0.749462481	0.453578504	1.000000000
## 76	Pop - Rock	-1.945897835	0.051667013	1.000000000
## 77	Rap - Rock	0.921857753	0.356602778	1.000000000
## 78	RnB - Rock	0.689546327	0.490479530	1.000000000
## 79	Classical - VGM	0.753774930	0.450984380	1.000000000
## 80	Country - VGM	1.213866649	0.224798650	1.000000000
## 81	EDM - VGM	2.428864307	0.015146200	1.000000000
## 82	Folk - VGM	1.331678122	0.182965982	1.000000000
## 83	HipHop - VGM	2.172326871	0.029831017	1.000000000
## 84	Jazz - VGM	3.243286709	0.001181592	0.10634332
## 85	Kpop - VGM	1.335954483	0.181564151	1.000000000
## 86	Metal - VGM	2.252698485	0.024278167	1.000000000
## 87	Other - VGM	1.784553326	0.074333798	1.000000000
## 88	Pop - VGM	0.469146904	0.638964635	1.000000000
## 89	Rap - VGM	1.998476979	0.045664973	1.000000000
## 90	RnB - VGM	1.946150921	0.051636613	1.000000000
## 91	Rock - VGM	1.864976018	0.062184766	1.000000000

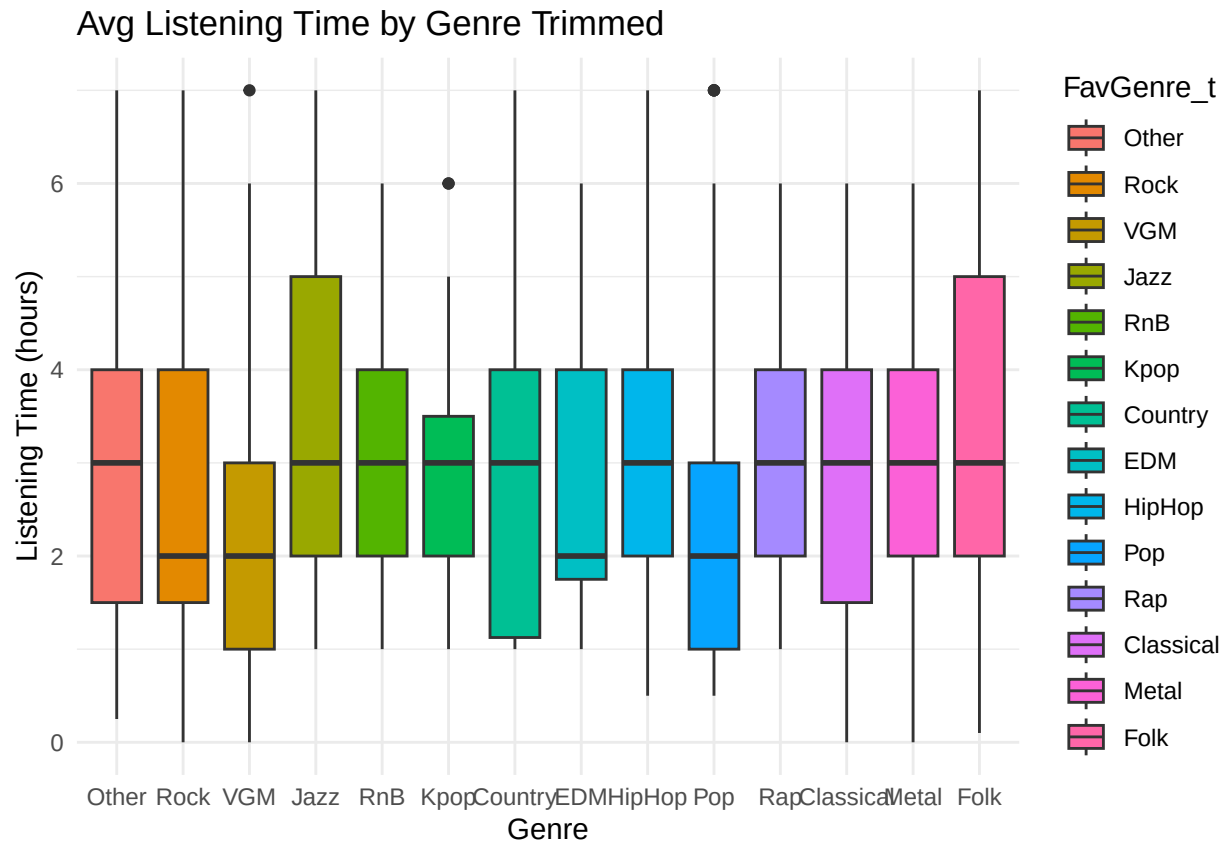
Rezultat pokazuje statistički značajnu razliku jedino između Jazz i pop žanrova na razini značajnosti od 10% sa P.adj vrijednosti od 0,099, te Jazz i Video game music skoro prelaze prag sa P.adj od 0,106.

Prikaz podataka u box dijagramima

```
ggplot(data, aes(x = FavGenre, y = AvgTime, fill = FavGenre)) +
  geom_boxplot() +
  theme_minimal() +
  labs(title = "Avg Listening Time by Genre", y = "Listening Time (hours)", x = "Genre")
```

```
ggplot(trimmed_data, aes(x = FavGenre_t, y = AvgTime_t, fill = FavGenre_t)) +
  geom_boxplot() +
  theme_minimal() +
  labs(title = "Avg Listening Time by Genre Trimmed", y = "Listening Time (hours)", x = "Genre")
```



5. Predviđanje dobi ispitanika

Naš zadatak je pokušati pomoću podataka predvidjeti dob ispitanika.

Nekategorijske varijable

Prvo ćemo promatrati sljedeće nekategorijske varijable:

```
names(ver2)[sapply(ver2, is.numeric)]
```

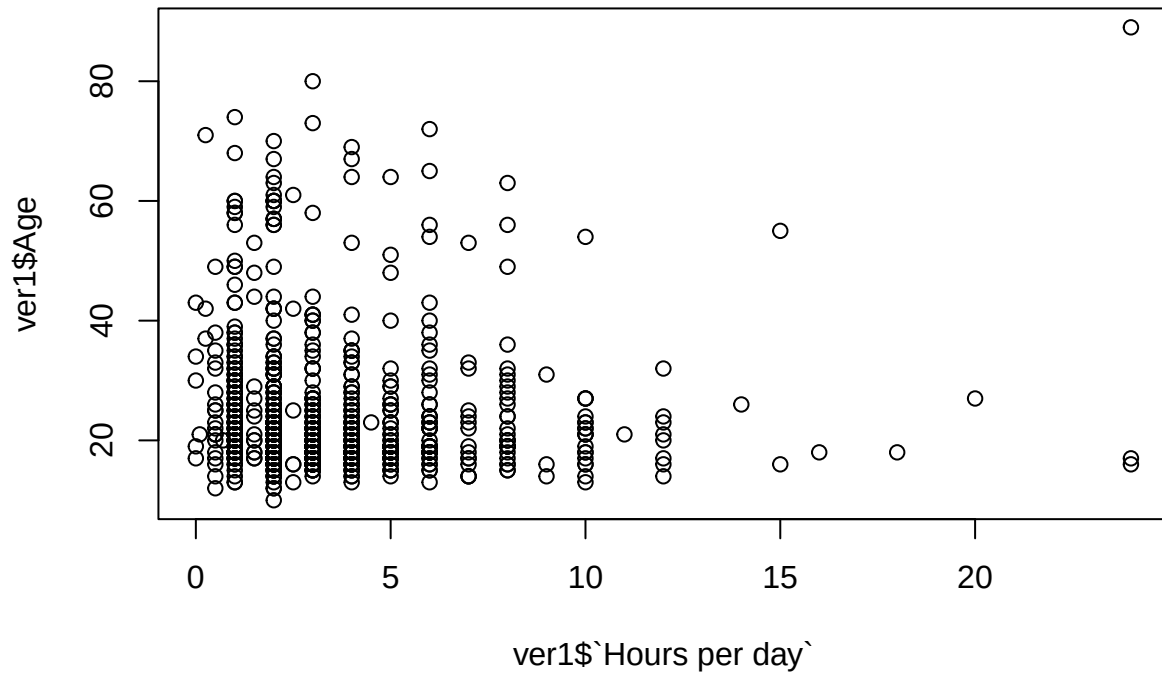
```
## [1] "Age" "Hours per day" "While working"
## [4] "Instrumentalist" "Composer" "Exploratory"
## [7] "Foreign languages" "Anxiety" "Depression"
## [10] "Insomnia" "OCD" "Music effects"
```

Kasnije ćemo se baviti frekvencijama slušanja.

Idemo provjeriti možemo li nešto naslutiti iz grafova.

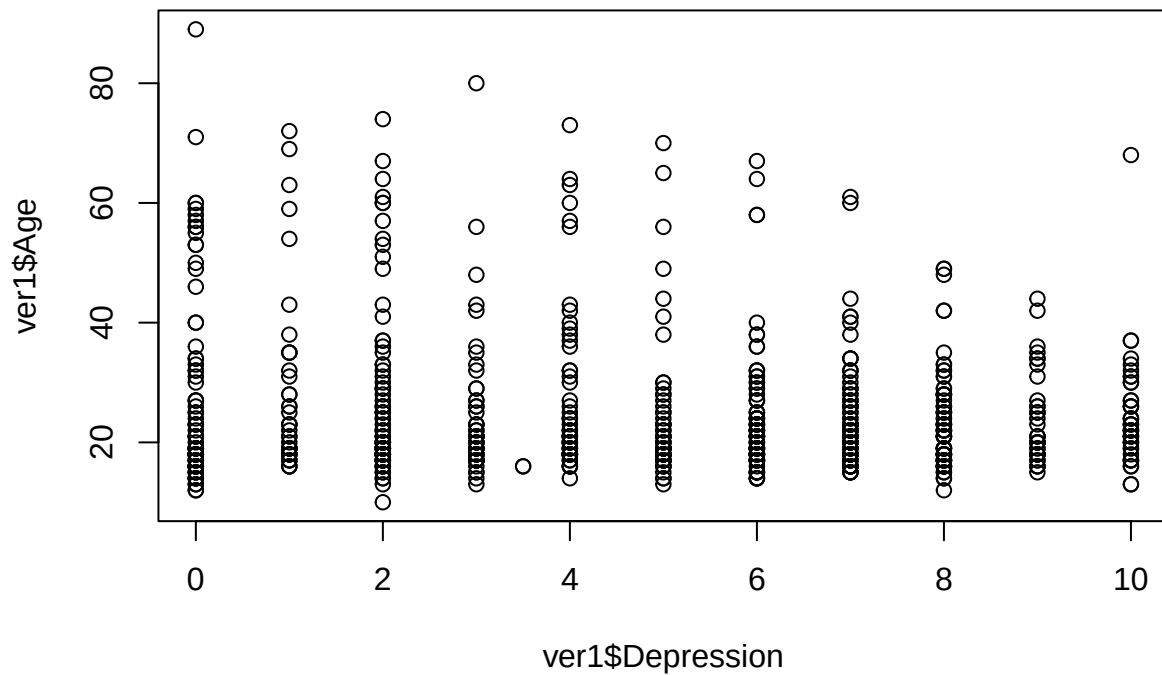
```
plot(ver1$`Hours per day`, ver1$Age, main = "Scatter plot for Age ~ Hours per day")
```

Scatter plot for Age ~ Hours per day



```
plot(ver1$`Depression`, ver1$Age, main = "Scatter plot for Age ~ Depression")
```

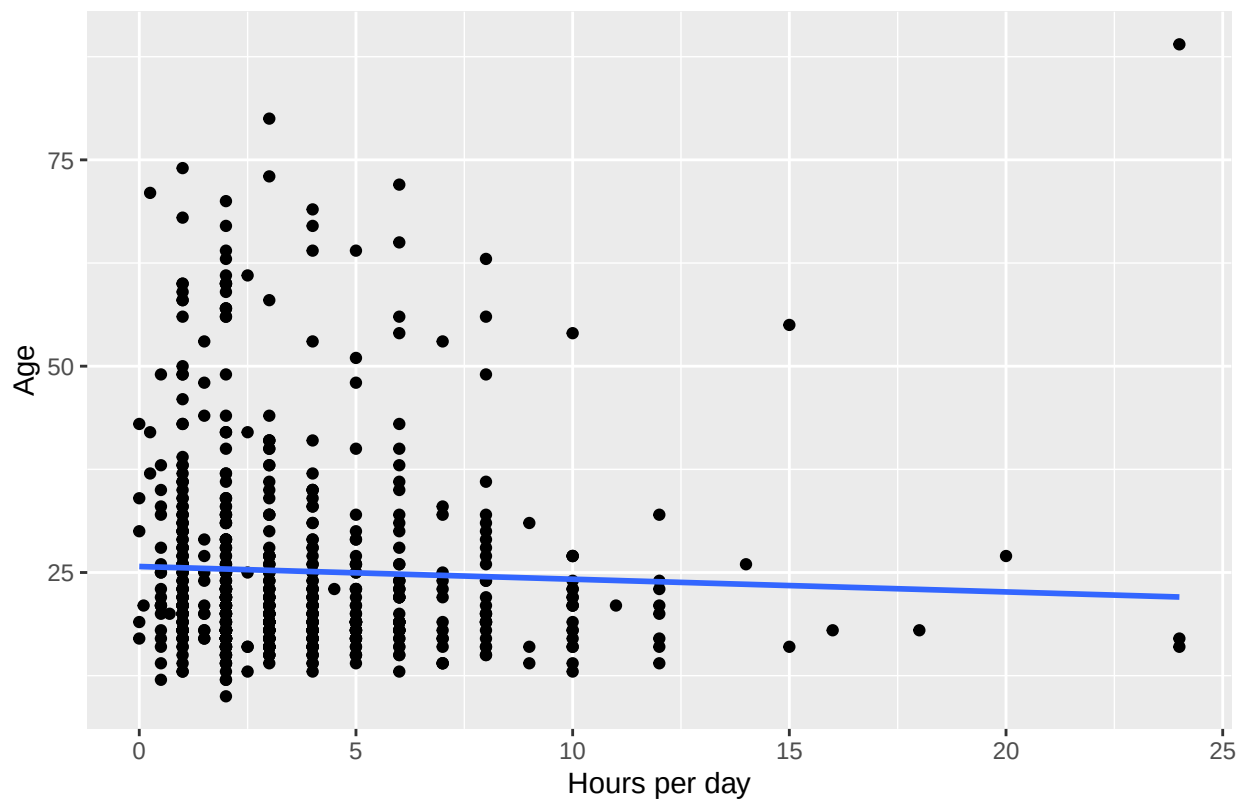
Scatter plot for Age ~ Depression



Iz grafova se ne može primijetiti neka linearna ovisnost broja godina o nekoj karakteristici. No, svejedno ćemo pokušati napraviti linearne modele varijabli i provjeriti rezultate.

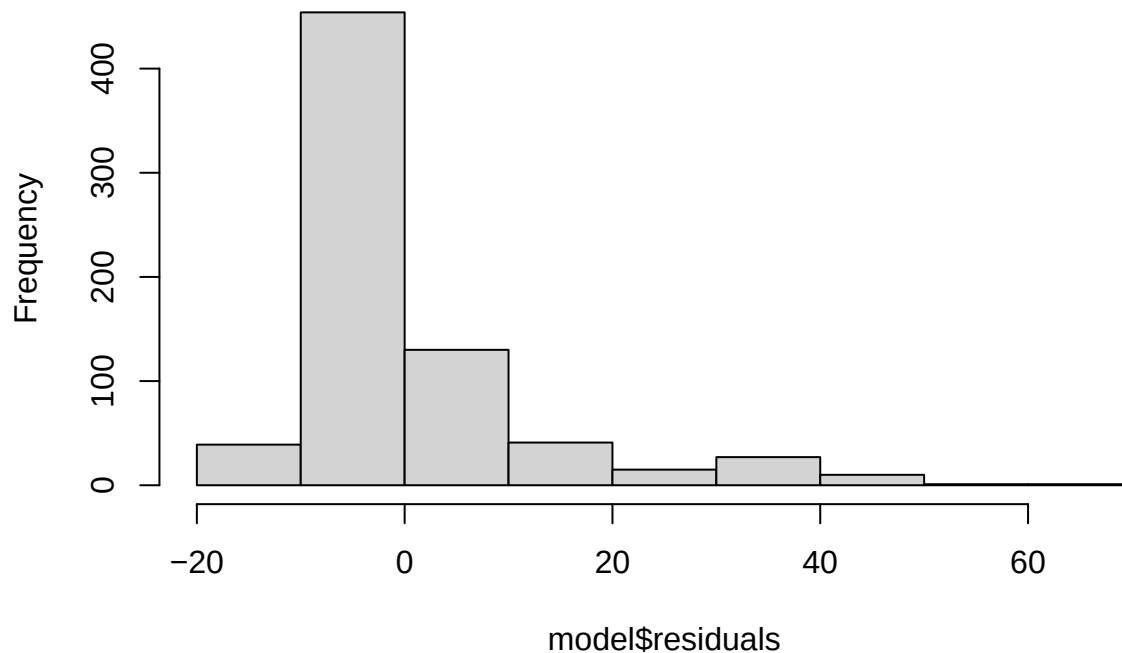
```
ggplot(ver1, aes(x = `Hours per day` , y = Age)) + geom_point() + stat_smooth(method = "lm", se = FALSE,
  title = "Scatter plot with Linear Regression Line for Hours per day",
  x = "Hours per day",
  y = "Age"
)
```

Scatter plot with Linear Regression Line for Hours per day



```
model <- lm(Age ~ `Hours per day`, data = ver1)
summary(model)
hist(model$residuals, main = "Histogram za rezidualne Age ~ Hours per day")
```

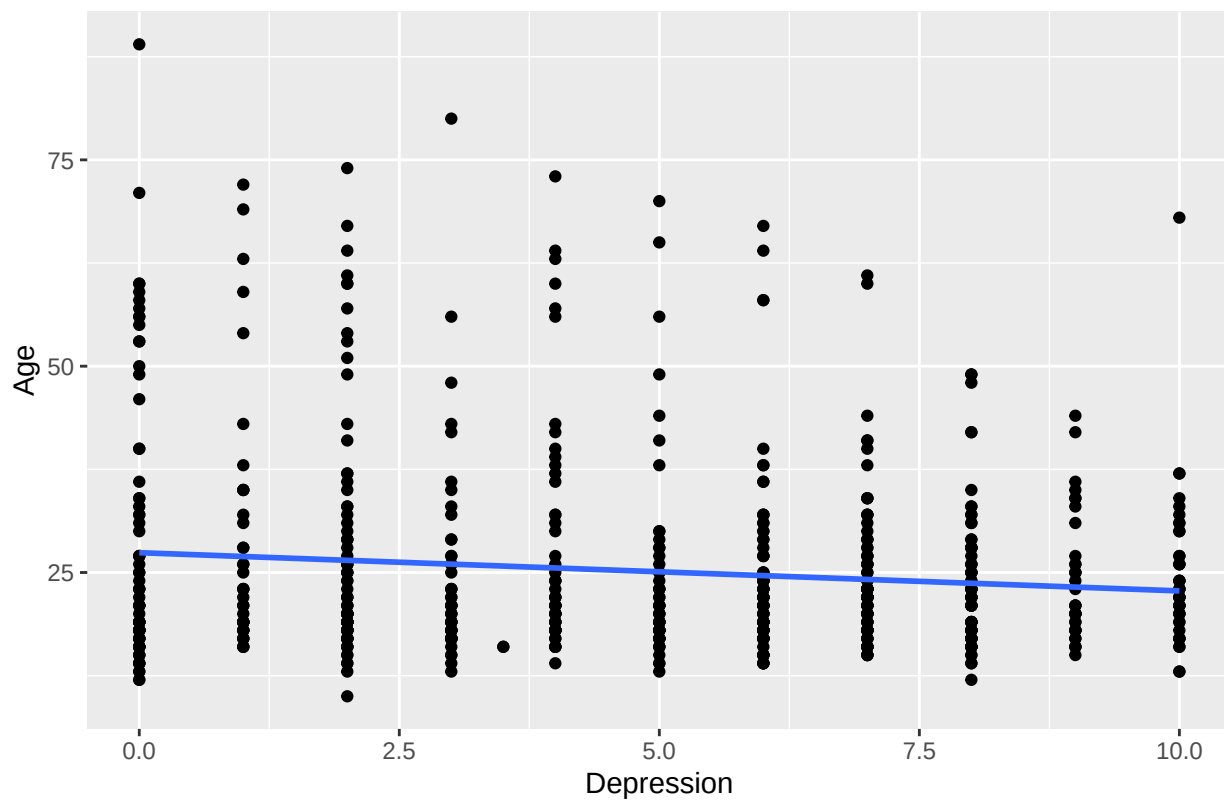
Histogram za reziduale Age ~ Hours per day



```
##
## Call:
## lm(formula = Age ~ 'Hours per day', data = ver1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.420  -7.420  -4.266   2.561  66.968
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    25.7279     0.6922  37.169  <2e-16 ***
## 'Hours per day' -0.1540     0.1479  -1.041    0.298
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.95 on 716 degrees of freedom
## Multiple R-squared:  0.001511,    Adjusted R-squared:  0.0001169
## F-statistic: 1.084 on 1 and 716 DF,  p-value: 0.2982
```

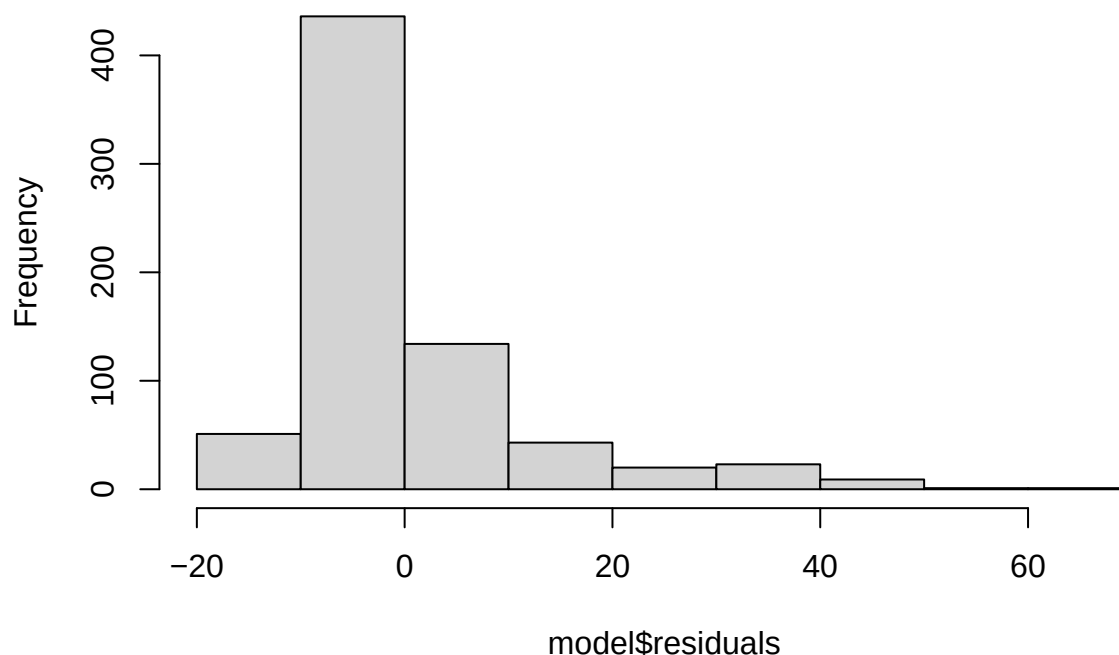
```
ggplot(ver1, aes(x = `Depression` , y = Age)) + geom_point() + stat_smooth(method = "lm", se = FALSE, f
  title = "Scatter plot with Linear Regression Line for Depression",
  x = "Depression",
  y = "Age"
)
```

Scatter plot with Linear Regression Line for Depression



```
model <- lm(Age ~ `Depression`, data = ver1)
summary(model)
hist(model$residuals, main = "Histogram za rezidualne Age ~ Depression")
```

Histogram za reziduale Age ~ Depression



```
##
## Call:
## lm(formula = Age ~ Depression, data = ver1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -16.477  -7.550  -3.697   2.913  61.597
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  27.4034     0.8338  32.868 < 2e-16 ***
## Depression   -0.4632     0.1469  -3.153  0.00168 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.88 on 716 degrees of freedom
## Multiple R-squared:  0.0137, Adjusted R-squared:  0.01232
## F-statistic: 9.943 on 1 and 716 DF, p-value: 0.001682
```

Nažalost, modeli potvrđuju da se iz njih ne može iščitati neka linearna povezanost, štoviše, neke varijable čak imaju p-vrijednost veću od 0.05 te na značajnosti od $\alpha = 0.05$ možemo reći da ne doprinose u predviđanju dobi ispitanika. Osim toga, reziduali modela se moraju ravnati po normalnoj razdiobi tako da bismo još morali provesti test normalnosti za reziduale.


```
lillie.test(rstandard(model))
```

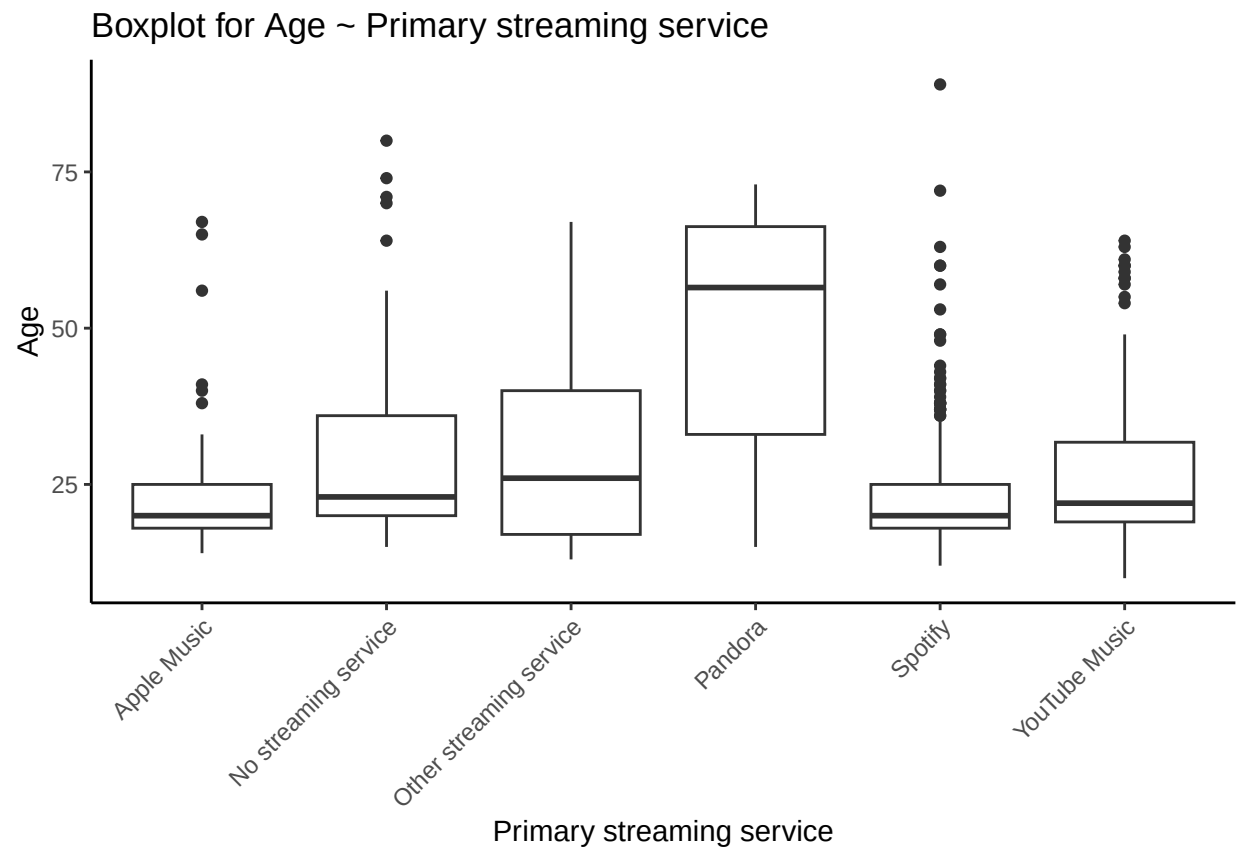
```
##  
## Lilliefors (Kolmogorov-Smirnov) normality test  
##  
## data:  rstandard(model)  
## D = 0.1821, p-value < 2.2e-16
```

Vidimo da reziduali modela ne prolaze test normalnosti, što je još jedan problem.

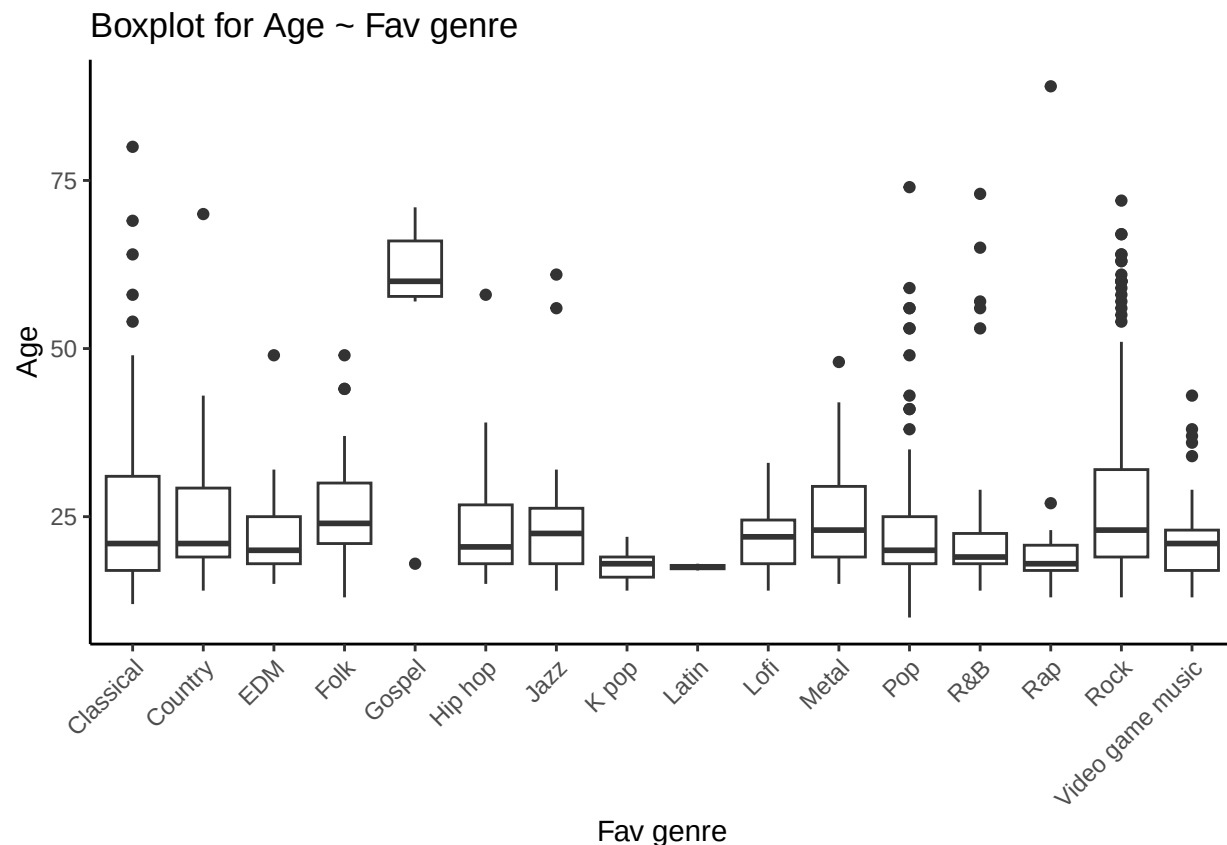
Kategorijske varijable

Idemo provjeriti ima li veza između broja godina i kategorijskih varijabli. Kategorijske varijable su *Primary streaming service* i *Fav genre*.

```
ver1$`Primary streaming service` <- as.factor(ver1$`Primary streaming service`)  
ver1$`Fav genre` <- as.factor(ver1$`Fav genre`)  
ver1$Timestamp <- NULL  
  
ver2$`Primary streaming service` <- as.factor(ver2$`Primary streaming service`)  
ver2$`Fav genre` <- as.factor(ver2$`Fav genre`)  
ver2$Timestamp <- NULL  
ver2 <- ver2 %>%  
  mutate(across(starts_with("Frequency"), ~ factor(., levels = c("Never", "Rarely", "Sometimes", "Very  
often"))))  
  
ggplot(ver1, aes(x = `Primary streaming service`, y = Age)) + geom_boxplot() + labs(title = "Boxplot for  
Primary streaming service")
```



```
ggplot(ver1, aes(x = `Fav genre`, y = Age)) + geom_boxplot() + labs(title = "Boxplot for Age ~ Fav genre")
```



Nažalost, niti grafovi kategorijskih varijabli ne daju nam konkretan odgovor. Svejedno, pokušat ćemo napraviti linearni model obje kategorijske variјable.

```
model <- lm(Age ~ `Primary streaming service`, data = ver1)
```

```
summary(model)
```

```
##
## Call:
## lm(formula = Age ~ 'Primary streaming service', data = ver1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -33.900  -6.718  -3.718   3.282  66.282
##
## Coefficients:
##                                Estimate Std. Error t value
## (Intercept)                   24.280     1.587    15.299
## 'Primary streaming service'No streaming service      6.662     2.084     3.196
## 'Primary streaming service'Other streaming service    6.189     2.256     2.744
## 'Primary streaming service'Pandora                  24.620     3.887     6.333
## 'Primary streaming service'Spotify                   -1.562     1.673    -0.934
## 'Primary streaming service'YouTube Music              3.753     1.979     1.896
##                                Pr(>|t|)
## (Intercept)                                < 2e-16 ***
## 'Primary streaming service'No streaming service    0.00145 **
```

```
## 'Primary streaming service'Other streaming service 0.00623 **
## 'Primary streaming service'Pandora 4.25e-10 ***
## 'Primary streaming service'Spotify 0.35070
## 'Primary streaming service'YouTube Music 0.05834 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.22 on 712 degrees of freedom
## Multiple R-squared:  0.1248, Adjusted R-squared:  0.1187
## F-statistic: 20.31 on 5 and 712 DF,  p-value: < 2.2e-16
```

```
model <- lm(Age ~ `Fav genre`, data = ver1)
```

```
summary(model)
```

```
##
## Call:
## lm(formula = Age ~ 'Fav genre', data = ver1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -37.667  -6.959  -3.254   2.861   67.227
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      26.726      1.605  16.652 < 2e-16 ***
## 'Fav genre'Country      -1.184      2.837   -0.417  0.67663
## 'Fav genre'EDM          -4.642      2.495   -1.861  0.06322 .
## 'Fav genre'Folk         -0.760      2.666   -0.285  0.77566
## 'Fav genre'Gospel      28.941      4.947   5.850 7.52e-09 ***
## 'Fav genre'Hip hop     -3.284      2.538   -1.294  0.19602
## 'Fav genre'Jazz        -1.276      3.024   -0.422  0.67331
## 'Fav genre'K pop       -8.964      2.972   -3.016  0.00265 **
## 'Fav genre'Latin       -9.226      8.262   -1.117  0.26455
## 'Fav genre'Lofi        -4.625      3.964   -1.167  0.24366
## 'Fav genre'Metal       -1.679      2.021   -0.831  0.40633
## 'Fav genre'Pop         -3.471      1.931   -1.798  0.07266 .
## 'Fav genre'R&B         -1.383      2.516   -0.550  0.58279
## 'Fav genre'Rap         -4.953      2.924   -1.694  0.09070 .
## 'Fav genre'Rock         1.296      1.814    0.715  0.47506
## 'Fav genre'Video game music -5.051      2.373   -2.129  0.03364 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.46 on 702 degrees of freedom
## Multiple R-squared:  0.09987, Adjusted R-squared:  0.08064
## F-statistic: 5.192 on 15 and 702 DF,  p-value: 6.861e-10
```

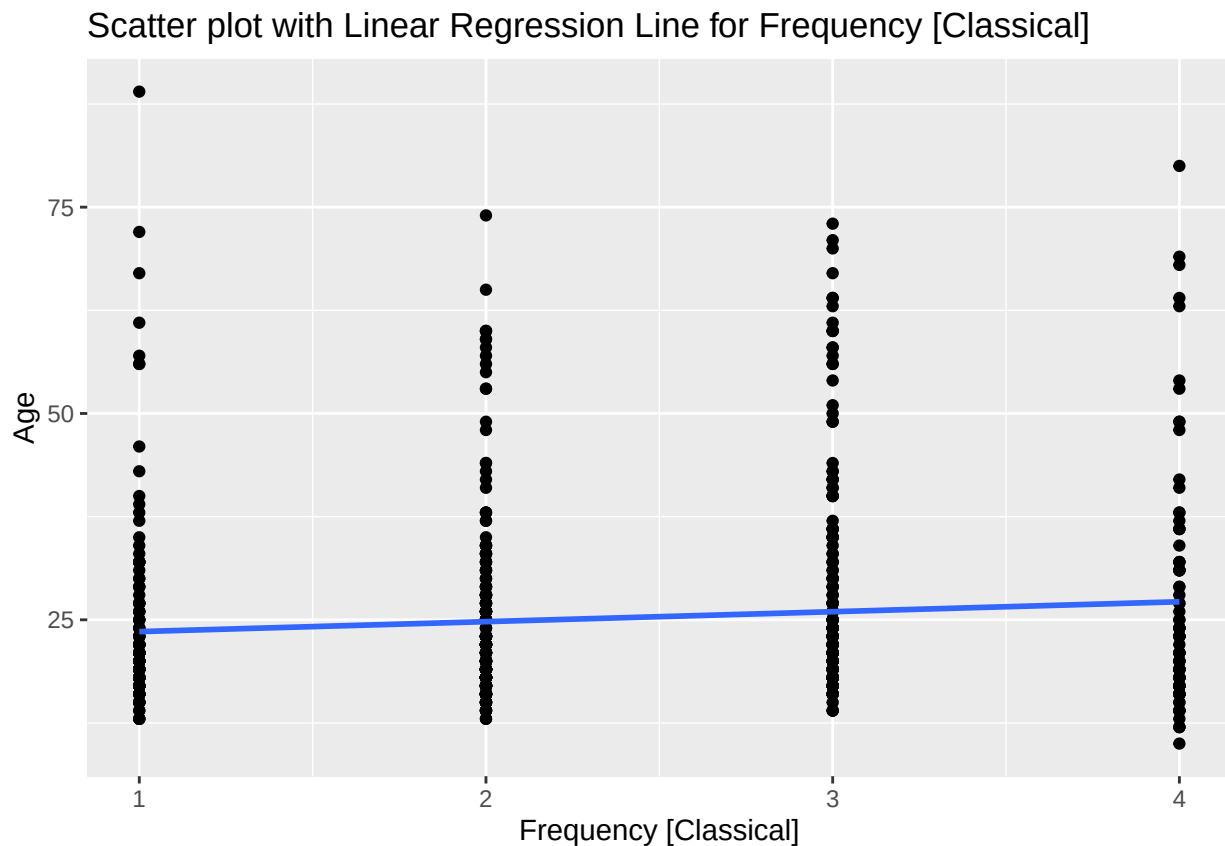
R-kvadrat vrijednosti ova dva modela su vrlo niske. Sve je jasnije da ćemo teško moći doći do kvalitetnog rezultata.

Frekvencije slušanja

Frekvencije slušanja promatrat ćemo i kao nekategorijske varijable i kao kategorijske i onda izabrati ono što je ostvarilo bolje rezultate.

Prvo ćemo ih promatrati kao nekategorijske varijable. U ovom slučaju kategorije *Never*, *Rarely*, *Sometimes* i *Very frequently* gledat ćemo kao brojeve 1, 2, 3 i 4. Napraviti ćemo linearne modele za nekoliko žanrova.

```
ggplot(ver1, aes(x = `Frequency [Classical]`, y = Age)) + geom_point() + stat_smooth(method = "lm", se
  title = "Scatter plot with Linear Regression Line for Frequency [Classical]",
  x = "Frequency [Classical]",
  y = "Age"
)
```



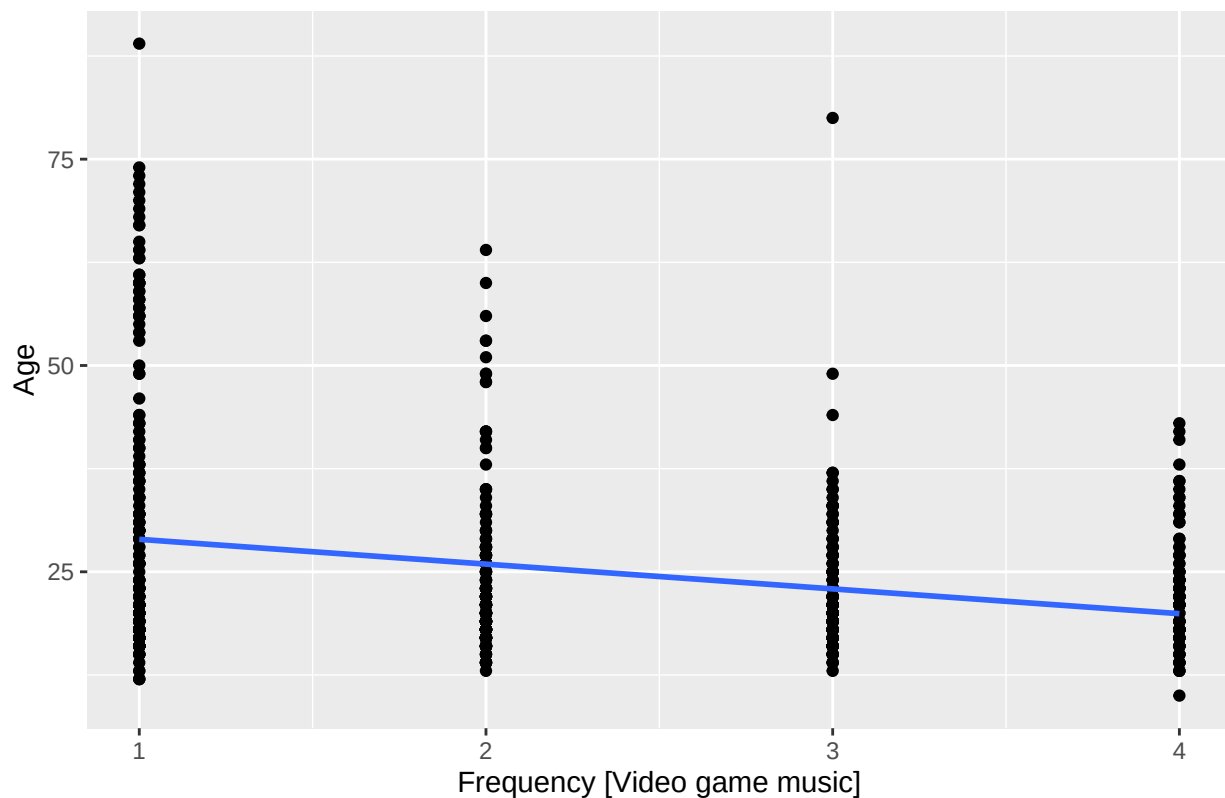
```
model <- lm(Age ~ `Frequency [Classical]`, data = ver1)
summary(model)
```

```
##
## Call:
## lm(formula = Age ~ `Frequency [Classical]`, data = ver1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -17.186  -6.981  -3.776   3.019  65.429
```

```
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      22.3655      1.1373   19.666 < 2e-16 ***
## 'Frequency [Classical]'  1.2051      0.4488    2.685  0.00741 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.9 on 716 degrees of freedom
## Multiple R-squared:  0.009971, Adjusted R-squared:  0.008588
## F-statistic: 7.211 on 1 and 716 DF, p-value: 0.007413
```

```
ggplot(ver1, aes(x = `Frequency [Video game music]`, y = Age)) + geom_point() + stat_smooth(method = "lm",
  title = "Scatter plot with Linear Regression Line for Frequency [Video game music]",
  x = "Frequency [Video game music]",
  y = "Age"
)
```

Scatter plot with Linear Regression Line for Frequency [Video game music]



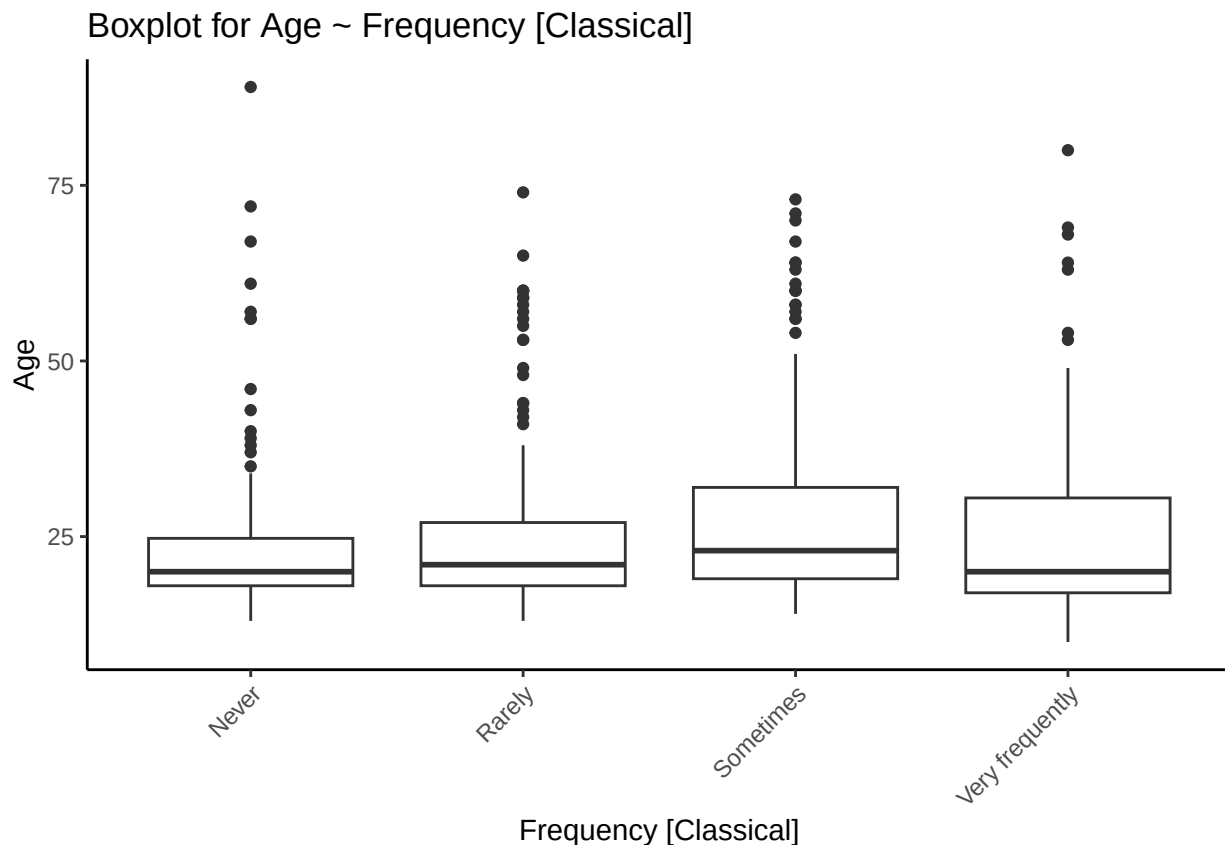
```
model <- lm(Age ~ `Frequency [Video game music]`, data = ver1)
summary(model)
```

```
##
## Call:
## lm(formula = Age ~ 'Frequency [Video game music]', data = ver1)
```

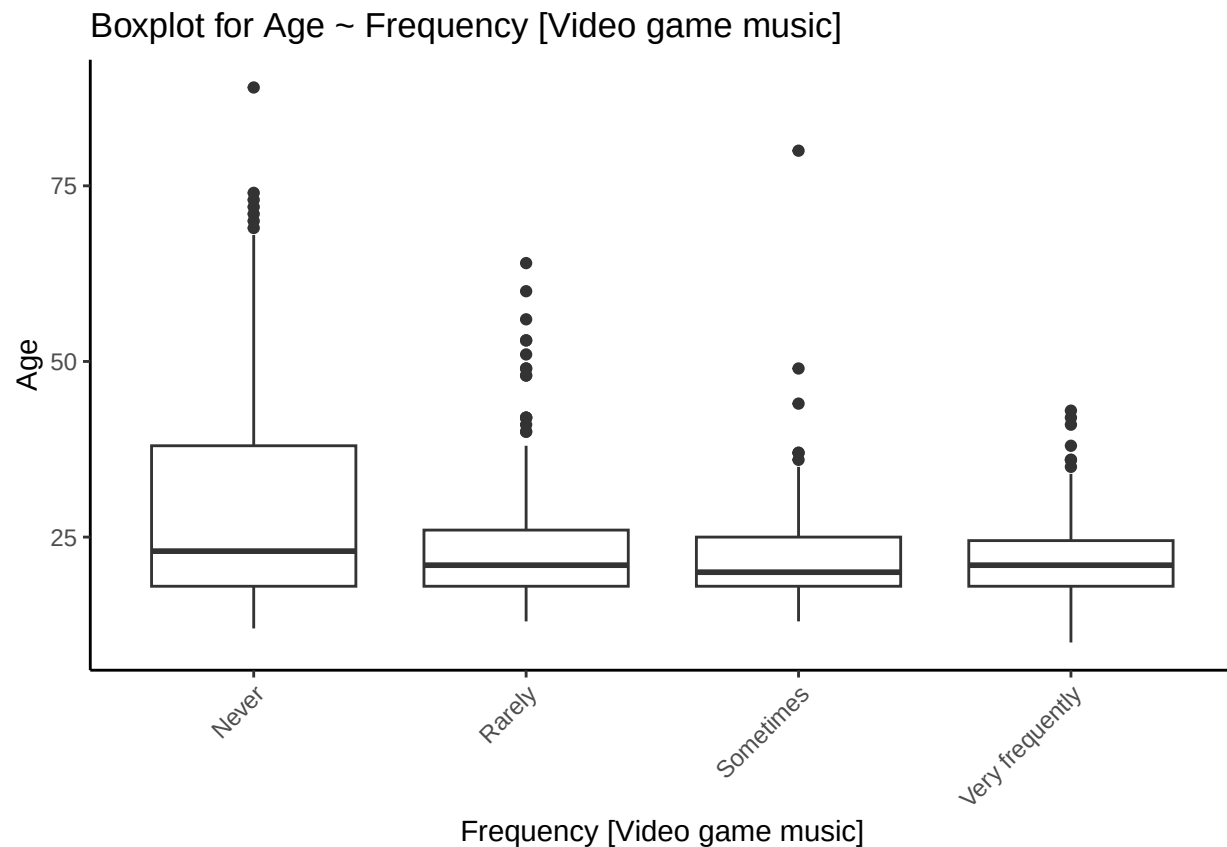
```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -16.936  -6.936  -2.934   3.069  60.064
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      31.9383     1.0012   31.899 < 2e-16 ***
## 'Frequency [Video game music]' -3.0023     0.4015  -7.478 2.22e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.52 on 716 degrees of freedom
## Multiple R-squared:  0.07244,    Adjusted R-squared:  0.07114
## F-statistic: 55.92 on 1 and 716 DF,  p-value: 2.217e-13
```

Opet ne ostvarujemo dobre rezultate. Sad ćemo ih promatrati kao kategorijske varijable. Za iste žanrove radit ćemo boxplotove.

```
ggplot(ver2, aes(x = `Frequency [Classical]`, y = Age)) + geom_boxplot() + labs(title = "Boxplot for Age
```



```
ggplot(ver2, aes(x = `Frequency [Video game music]`, y = Age)) + geom_boxplot() + labs(title = "Boxplot
```



```
model <- lm(Age ~ `Frequency [Classical]`, data = ver2)
```

```
summary(model)
```

```
##
## Call:
## lm(formula = Age ~ `Frequency [Classical]`, data = ver2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.292  -7.292  -4.259   2.683  65.741
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    23.2590     0.9202  25.275  < 2e-16 ***
## `Frequency [Classical]`Rarely      1.1323     1.1842   0.956  0.339339
## `Frequency [Classical]`Sometimes    4.5337     1.2551   3.612  0.000325 ***
## `Frequency [Classical]`Very frequently  2.0334     1.4741   1.379  0.168193
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.86 on 714 degrees of freedom
## Multiple R-squared:  0.02039,    Adjusted R-squared:  0.01627
## F-statistic: 4.953 on 3 and 714 DF,  p-value: 0.002075
```



```
model <- lm(Age ~ `Frequency [Video game music]`, data = ver2)

summary(model)
```

```
##
## Call:
## lm(formula = Age ~ 'Frequency [Video game music]', data = ver2)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-18.472	-6.593	-3.096	2.904	58.528

```
##
## Coefficients:
```

	Estimate	Std. Error	t value
(Intercept)	30.4716	0.7533	40.453
'Frequency [Video game music]'Rarely	-6.8788	1.1123	-6.184
'Frequency [Video game music]'Sometimes	-8.3549	1.1354	-7.358
'Frequency [Video game music]'Very frequently	-8.3760	1.3028	-6.429

```
##
## Pr(>|t|)
```

	Pr(> t)
(Intercept)	< 2e-16 ***
'Frequency [Video game music]'Rarely	1.05e-09 ***
'Frequency [Video game music]'Sometimes	5.13e-13 ***
'Frequency [Video game music]'Very frequently	2.35e-10 ***

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.4 on 714 degrees of freedom
## Multiple R-squared:  0.09452,    Adjusted R-squared:  0.09071
## F-statistic: 24.84 on 3 and 714 DF,  p-value: 2.682e-15
```

Bolje rezultate pokazuje linearni model gdje frekvencije slušanja promatramo kao kategorijske varijable, ali ne za preznačajnu razliku jer u oba slučaja R kvadrat vrijednost nije velika.

Konačni pokušaj modela linearne regresije

Usprkos svemu, provjeirćemo kakav linearni model ćemo dobiti kada sve (značajne) varijable skupimo u jedan model.

```
ver3 <- ver2 %>% select(-`Music effects`, -`Hours per day`, -`While working`, -`Foreign languages`, -`Correlation`)
```

```
model <- lm(Age ~ ., data = ver3)

summary(model)
```

```
##
## Call:
## lm(formula = Age ~ ., data = ver3)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-18.960	-5.769	-0.906	4.315	59.290

```

##
## Coefficients:
##
## (Intercept) 30.98379
## 'Primary streaming service'I do not use a streaming service. 3.37442
## 'Primary streaming service'Other streaming service 3.14480
## 'Primary streaming service'Pandora 14.34877
## 'Primary streaming service'Spotify -0.99044
## 'Primary streaming service'YouTube Music 1.34185
## Instrumentalist -3.24684
## 'Fav genre'Country -2.65529
## 'Fav genre'EDM -2.75489
## 'Fav genre'Folk 0.20021
## 'Fav genre'Gospel 15.97985
## 'Fav genre'Hip hop -1.64051
## 'Fav genre'Jazz -0.60188
## 'Fav genre'K pop -1.29042
## 'Fav genre'Latin -2.95196
## 'Fav genre'Lofi 0.89260
## 'Fav genre'Metal 2.63971
## 'Fav genre'Pop 0.03208
## 'Fav genre'R&B -0.47804
## 'Fav genre'Rap -0.97400
## 'Fav genre'Rock 3.19760
## 'Fav genre'Video game music -4.04180
## Exploratory -2.81998
## 'Frequency [Classical]'Rarely 0.69389
## 'Frequency [Classical]'Sometimes 4.01392
## 'Frequency [Classical]'Very frequently 1.83628
## 'Frequency [Country]'Rarely 0.83580
## 'Frequency [Country]'Sometimes 2.62708
## 'Frequency [Country]'Very frequently 2.27875
## 'Frequency [EDM]'Rarely -0.67789
## 'Frequency [EDM]'Sometimes -0.38979
## 'Frequency [EDM]'Very frequently 1.87807
## 'Frequency [Folk]'Rarely 1.81256
## 'Frequency [Folk]'Sometimes 2.66299
## 'Frequency [Folk]'Very frequently 1.39807
## 'Frequency [Gospel]'Rarely -1.46755
## 'Frequency [Gospel]'Sometimes -0.23565
## 'Frequency [Gospel]'Very frequently 0.07248
## 'Frequency [Hip hop]'Rarely 0.46459
## 'Frequency [Hip hop]'Sometimes 0.83022
## 'Frequency [Hip hop]'Very frequently 0.19922
## 'Frequency [Jazz]'Rarely 1.59615
## 'Frequency [Jazz]'Sometimes 0.76572
## 'Frequency [Jazz]'Very frequently 2.72350
## 'Frequency [K pop]'Rarely 1.47232
## 'Frequency [K pop]'Sometimes -0.43336
## 'Frequency [K pop]'Very frequently -2.05810
## 'Frequency [Latin]'Rarely 2.02411
## 'Frequency [Latin]'Sometimes 2.87776
## 'Frequency [Latin]'Very frequently -0.86567
## 'Frequency [Lofi]'Rarely -0.99469

```

## 'Frequency [Lofi]'Sometimes	-0.73175	
## 'Frequency [Lofi]'Very frequently	-2.81487	
## 'Frequency [Metal]'Rarely	0.83389	
## 'Frequency [Metal]'Sometimes	1.05928	
## 'Frequency [Metal]'Very frequently	-1.20716	
## 'Frequency [Pop]'Rarely	-0.39831	
## 'Frequency [Pop]'Sometimes	-2.15098	
## 'Frequency [Pop]'Very frequently	-5.43389	
## 'Frequency [R&B]'Rarely	1.68255	
## 'Frequency [R&B]'Sometimes	3.99165	
## 'Frequency [R&B]'Very frequently	4.64998	
## 'Frequency [Rap]'Rarely	-3.04364	
## 'Frequency [Rap]'Sometimes	-4.33026	
## 'Frequency [Rap]'Very frequently	-3.89289	
## 'Frequency [Rock]'Rarely	2.29790	
## 'Frequency [Rock]'Sometimes	3.66694	
## 'Frequency [Rock]'Very frequently	0.57307	
## 'Frequency [Video game music]'Rarely	-5.45334	
## 'Frequency [Video game music]'Sometimes	-6.74237	
## 'Frequency [Video game music]'Very frequently	-5.25942	
## Anxiety	-0.50119	
## Depression	-0.03955	
## Insomnia	0.23725	
## OCD	-0.24287	
##	Std. Error	t value
## (Intercept)	3.08508	10.043
## 'Primary streaming service'I do not use a streaming service.	1.92871	1.750
## 'Primary streaming service'Other streaming service	2.03185	1.548
## 'Primary streaming service'Pandora	3.63750	3.945
## 'Primary streaming service'Spotify	1.49452	-0.663
## 'Primary streaming service'YouTube Music	1.78358	0.752
## Instrumentalist	0.88033	-3.688
## 'Fav genre'Country	3.54089	-0.750
## 'Fav genre'EDM	2.83794	-0.971
## 'Fav genre'Folk	2.94056	0.068
## 'Fav genre'Gospel	5.33491	2.995
## 'Fav genre'Hip hop	2.86298	-0.573
## 'Fav genre'Jazz	3.22081	-0.187
## 'Fav genre'K pop	3.27915	-0.394
## 'Fav genre'Latin	7.37385	-0.400
## 'Fav genre'Lofi	3.79896	0.235
## 'Fav genre'Metal	2.54942	1.035
## 'Fav genre'Pop	2.24541	0.014
## 'Fav genre'R&B	2.74853	-0.174
## 'Fav genre'Rap	3.01167	-0.323
## 'Fav genre'Rock	2.15445	1.484
## 'Fav genre'Video game music	2.56626	-1.575
## Exploratory	0.91585	-3.079
## 'Frequency [Classical]'Rarely	1.03789	0.669
## 'Frequency [Classical]'Sometimes	1.12461	3.569
## 'Frequency [Classical]'Very frequently	1.63446	1.123
## 'Frequency [Country]'Rarely	0.92531	0.903
## 'Frequency [Country]'Sometimes	1.19158	2.205
## 'Frequency [Country]'Very frequently	2.28504	0.997

## 'Frequency [EDM] 'Rarely	0.98492	-0.688
## 'Frequency [EDM] 'Sometimes	1.09785	-0.355
## 'Frequency [EDM] 'Very frequently	1.58696	1.183
## 'Frequency [Folk] 'Rarely	0.98405	1.842
## 'Frequency [Folk] 'Sometimes	1.17683	2.263
## 'Frequency [Folk] 'Very frequently	1.58235	0.884
## 'Frequency [Gospel] 'Rarely	1.06286	-1.381
## 'Frequency [Gospel] 'Sometimes	1.53521	-0.153
## 'Frequency [Gospel] 'Very frequently	3.36732	0.022
## 'Frequency [Hip hop] 'Rarely	1.26719	0.367
## 'Frequency [Hip hop] 'Sometimes	1.49922	0.554
## 'Frequency [Hip hop] 'Very frequently	2.01185	0.099
## 'Frequency [Jazz] 'Rarely	0.95873	1.665
## 'Frequency [Jazz] 'Sometimes	1.13935	0.672
## 'Frequency [Jazz] 'Very frequently	1.94864	1.398
## 'Frequency [K pop] 'Rarely	0.99605	1.478
## 'Frequency [K pop] 'Sometimes	1.41688	-0.306
## 'Frequency [K pop] 'Very frequently	1.67542	-1.228
## 'Frequency [Latin] 'Rarely	1.00269	2.019
## 'Frequency [Latin] 'Sometimes	1.26032	2.283
## 'Frequency [Latin] 'Very frequently	2.04582	-0.423
## 'Frequency [Lofi] 'Rarely	0.99018	-1.005
## 'Frequency [Lofi] 'Sometimes	1.13131	-0.647
## 'Frequency [Lofi] 'Very frequently	1.46892	-1.916
## 'Frequency [Metal] 'Rarely	1.05527	0.790
## 'Frequency [Metal] 'Sometimes	1.25864	0.842
## 'Frequency [Metal] 'Very frequently	1.59163	-0.758
## 'Frequency [Pop] 'Rarely	1.63504	-0.244
## 'Frequency [Pop] 'Sometimes	1.61407	-1.333
## 'Frequency [Pop] 'Very frequently	1.73817	-3.126
## 'Frequency [R&B] 'Rarely	1.08325	1.553
## 'Frequency [R&B] 'Sometimes	1.22857	3.249
## 'Frequency [R&B] 'Very frequently	1.60338	2.900
## 'Frequency [Rap] 'Rarely	1.23869	-2.457
## 'Frequency [Rap] 'Sometimes	1.46889	-2.948
## 'Frequency [Rap] 'Very frequently	1.97642	-1.970
## 'Frequency [Rock] 'Rarely	1.54501	1.487
## 'Frequency [Rock] 'Sometimes	1.44635	2.535
## 'Frequency [Rock] 'Very frequently	1.58282	0.362
## 'Frequency [Video game music] 'Rarely	1.00310	-5.436
## 'Frequency [Video game music] 'Sometimes	1.06644	-6.322
## 'Frequency [Video game music] 'Very frequently	1.42552	-3.689
## Anxiety	0.16815	-2.981
## Depression	0.15561	-0.254
## Insomnia	0.13432	1.766
## OCD	0.13980	-1.737
##	Pr(> t)	
## (Intercept)	< 2e-16 ***	
## 'Primary streaming service'I do not use a streaming service.	0.080669 .	
## 'Primary streaming service'Other streaming service	0.122174	
## 'Primary streaming service'Pandora	8.87e-05 ***	
## 'Primary streaming service'Spotify	0.507748	
## 'Primary streaming service'YouTube Music	0.452124	
## Instrumentalist	0.000245 ***	

## 'Fav genre'Country	0.453593
## 'Fav genre'EDM	0.332045
## 'Fav genre'Folk	0.945739
## 'Fav genre'Gospel	0.002847 **
## 'Fav genre'Hip hop	0.566840
## 'Fav genre'Jazz	0.851819
## 'Fav genre'K pop	0.694065
## 'Fav genre'Latin	0.689048
## 'Fav genre'Lofi	0.814315
## 'Fav genre'Metal	0.300864
## 'Fav genre'Pop	0.988604
## 'Fav genre'R&B	0.861977
## 'Fav genre'Rap	0.746491
## 'Fav genre'Rock	0.138251
## 'Fav genre'Video game music	0.115753
## Exploratory	0.002165 **
## 'Frequency [Classical]'Rarely	0.504015
## 'Frequency [Classical]'Sometimes	0.000385 ***
## 'Frequency [Classical]'Very frequently	0.261654
## 'Frequency [Country]'Rarely	0.366726
## 'Frequency [Country]'Sometimes	0.027828 *
## 'Frequency [Country]'Very frequently	0.319018
## 'Frequency [EDM]'Rarely	0.491532
## 'Frequency [EDM]'Sometimes	0.722673
## 'Frequency [EDM]'Very frequently	0.237071
## 'Frequency [Folk]'Rarely	0.065945 .
## 'Frequency [Folk]'Sometimes	0.023978 *
## 'Frequency [Folk]'Very frequently	0.377276
## 'Frequency [Gospel]'Rarely	0.167831
## 'Frequency [Gospel]'Sometimes	0.878056
## 'Frequency [Gospel]'Very frequently	0.982834
## 'Frequency [Hip hop]'Rarely	0.714014
## 'Frequency [Hip hop]'Sometimes	0.579931
## 'Frequency [Hip hop]'Very frequently	0.921149
## 'Frequency [Jazz]'Rarely	0.096429 .
## 'Frequency [Jazz]'Sometimes	0.501782
## 'Frequency [Jazz]'Very frequently	0.162703
## 'Frequency [K pop]'Rarely	0.139855
## 'Frequency [K pop]'Sometimes	0.759816
## 'Frequency [K pop]'Very frequently	0.219743
## 'Frequency [Latin]'Rarely	0.043935 *
## 'Frequency [Latin]'Sometimes	0.022735 *
## 'Frequency [Latin]'Very frequently	0.672332
## 'Frequency [Lofi]'Rarely	0.315489
## 'Frequency [Lofi]'Sometimes	0.517983
## 'Frequency [Lofi]'Very frequently	0.055772 .
## 'Frequency [Metal]'Rarely	0.429696
## 'Frequency [Metal]'Sometimes	0.400320
## 'Frequency [Metal]'Very frequently	0.448463
## 'Frequency [Pop]'Rarely	0.807612
## 'Frequency [Pop]'Sometimes	0.183120
## 'Frequency [Pop]'Very frequently	0.001851 **
## 'Frequency [R&B]'Rarely	0.120856
## 'Frequency [R&B]'Sometimes	0.001218 **

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## 'Frequency [R&B]'Very frequently      0.003858 **
## 'Frequency [Rap]'Rarely                 0.014267 *
## 'Frequency [Rap]'Sometimes              0.003315 **
## 'Frequency [Rap]'Very frequently        0.049305 *
## 'Frequency [Rock]'Rarely                0.137424
## 'Frequency [Rock]'Sometimes             0.011471 *
## 'Frequency [Rock]'Very frequently       0.717431
## 'Frequency [Video game music]'Rarely    7.73e-08 ***
## 'Frequency [Video game music]'Sometimes 4.82e-10 ***
## 'Frequency [Video game music]'Very frequently 0.000244 ***
## Anxiety                                0.002984 **
## Depression                             0.799449
## Insomnia                               0.077817 .
## OCD                                    0.082810 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.476 on 643 degrees of freedom
## Multiple R-squared:  0.4365, Adjusted R-squared:  0.3716
## F-statistic:  6.73 on 74 and 643 DF,  p-value: < 2.2e-16

```

Dobili smo model koji ima R-kvadrat vrijednost 0.372, veću nego kada gledamo svaku varijablu zasebno, no opet nedovoljno veliku da možemo zaključiti da na temelju ovog modela možemo predvidjeti dob ispitanika. Taj rezultat je očekivan iz razloga što svaki čovjek je različit i ne postoji određena pravilnost po kojoj možemo točno predvidjeti koliko godina neka osoba ima. Općenito, stvari koje imaju veze s ljudima i ljudskim karakteristikama teže je predvidjeti zbog ljudskog nepredvidljivog ponašanja.